DiffDual-AD: Diffusion-Based Dual-Stage Adversarial Defense Framework in Remote Sensing With Denoiser Constraint

Zihao Lu, Hao Sun, Lin Lei, Yanjie Xu, Yuli Sun, and Gangyao Kuang, *Senior Member, IEEE*

***Abstract*— Deep neural networks (DNNs), though highly effective in various Earth observation tasks with remote sensing images (RSIs), are vulnerable to adversarial attacks, threatening their reliability. Each maliciously attacked RSI potentially contains unique and critical information, but current defenses lack a unified framework for rapidly detecting adversarial RSIs and accurately restoring them to their natural state. To bridge this research gap, we propose a diffusion-based dual-stage adversarial defense (DiffDual-AD) framework. In the first stage, we propose a novel adversarial detection method based on the score expectation (AD-SE), which is integrated into the forward process of diffusion model with an improved denoiser constrained for adversarial defense. During the reverse process, the second stage introduces the distance and label-guided adversarial purification (DL-GAP) to restore adversarial RSIs to natural ones. With the help of two proposed guidelines and the specific denoiser, the DL-GAP effectively smooths out the adversarial perturbations from the detected adversarial RSIs while preserving semantic information and local key features. Finally, DL-GAP yields non-adversarial purified RSIs based on the results of AD-SE. Both stages are integrated within the diffusion models, complementing each other to form a pipelined operational mode. Extensive experiments across three RSI scene classification datasets have proven its efficacy in resisting adversarial RSIs in both transparent-box and opaque-box scenarios, achieving an average adversarial detection accuracy of 87.02% with AD-SE and an average final classification accuracy of 81.50% with the entire DiffDual-AD. The performances of our proposed AD-SE and DL-GAP have surpassed other advanced adversarial detection and purification methods when applied to RSIs. Therefore, DiffDual-AD provides a unified and universal solution to preserve the utility and security of processing the adversarial RSIs, which advances the adversarial defense research in remote sensing.**

***Index Terms*— Adversarial detection, adversarial purification, diffusion models, dual stage, model robustness.**

I. Introduction

DEEP neural networks (DNNs) have demonstrated remarkable capabilities in feature extraction and representation learning of high-resolution remote sensing images (RSIs) [1], [2], [3], achieving state-of-the-art performance in various geoscience and remote sensing tasks for military (e.g., battlefield reconnaissance [4] and target detection [5], [6]) and civilian (e.g., land use classification [7], [8], postdisaster rescue [9], and change detection [10]) use. Despite acquiring notable success, high-performing DNNs have shown vulnerability to adversarial samples [11], [12], [13], which are generated by applying elaborately designed but imperceptible perturbations to natural data for degrading the performance of DNNs [14]. In remote sensing applications, adversarial samples also pose a significant threat. For example, as depicted in Fig. 1, a DNN-equipped drone that recognizes high-valued targets can be compromised when adversaries sneak into communication channels (e.g., Wi-Fi) or implant backdoor software for captured RSIs in real time. These RSIs are then efficiently crafted into adversarial samples using attack algorithms, leading to incorrect predictions from well-trained models [15]. Moreover, the characteristic of adversarial transferability among different DNNs further reduces the difficulty of mounting an adversarial attack [16].

Under the threat of adversarial samples in security-sensitive remote sensing areas, it is imperative to develop defense methods and improve the adversarial robustness of DNNs. Existing defense strategies are broadly categorized into three types: adversarial training (AT), detection, and purification. Although these mitigate vulnerabilities to some extent, in remote sensing, each maliciously attacked RSI may contain unique and critical information, and there is currently a lack of a unified framework that integrates rapidly detecting adversarial RSIs with accurately restoring them to their original state for further interpretation. The individual deployment of existing defense strategies cannot achieve this due to their respective inherent limitations.

Among the defense strategies, AT was the first to be studied for defending against adversarial RSIs. Standard AT forces DNNs to learn robust features and minimizes loss with adversarial samples generated online [17], [18]. However, it only defends against the specific attacks that they are trained with, leading to an obvious performance drop in handling unknown adversarial samples and even natural data [19]. Despite many improved versions like those in [20] and [21] for remote sensing scenarios, the retraining pattern with high computational cost and inability to obtain natural RSIs does not make AT conducive to practical remote sensing applications.

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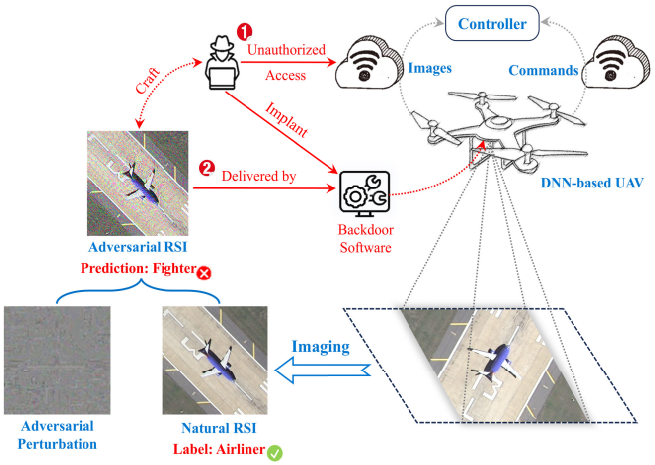


Fig. 1. One possible scenario for adversaries to manipulate captured imagery by infiltrating communication channels or embedding backdoor software. Adversarial RSIs lead to misclassification of targets, highlighting the urgent need for enhanced robustness of DNNs in remote sensing applications.

In contrast, adversarial detection does not modify the deployed DNNs, but rather solves the binary problem of whether an input image is adversarial or not, for which the key is to identify the discernible discrepancy between natural and adversarial distributions [22]. According to the detection results, natural samples are passed to the baseline DNNs, while adversarial samples are rejected for security [23], [24]. In the remote sensing field, the constant dynamics of natural and human activities on Earth's surface render each RSI acquisition unique and valuable. Therefore, detection-only defenses face limitations, primarily due to the potential for critical information loss when discarding the detected adversarial RSIs [25], [26].

Another defense strategy, known as adversarial purification, trains generative models separately from baseline DNNs to remove adversarial perturbations and restore attacked RSIs to their natural state [27], [28], [29], [30]. Early attempts include using denoising DNNs [27], [28] and generative neural networks (GANs) [29], [30] to purify adversarial RSIs. However, some common issues from these methods, such as the training instability and mode collapse of GAN models [31], as well as the inadequate generalization capability of denoising DNNs [32], hinder the performance improvement of adversarial purification. Recently, diffusion models, which have emerged as a potential solution to adversarial purification due to their strong sample quality and diversity [33], [34], can embed the purification strategy into two complementary processes: 1) a forward process that gradually injects Gaussian noise into an adversarial sample to submerge adversarial perturbations and 2) a reverse process that removes adversarial perturbations and Gaussian noise by denoising one step at a time [35], [36]. Leveraging the stochastic nature of both processes, the diffusion model can effectively restore the nonadversarial images while introducing controlled variations [37].

Despite these advantages, applying adversarial purification with the diffusion model to RSI interpretation still presents some challenges.

1) The reverse process involves denoising a test RSI sequentially from time step t to $t-1$, resulting in excessive response latency and computational costs [36]. Purifying all captured RSIs in real time, regardless of their adversarial status, is considered impractical.
2) The forward process is underexploited in existing adversarial purification methods.
3) The efficacy of purification heavily depends on the balance between completely removing perturbations and accurately preserving semantic content, particularly for RSIs characterized by intricate backgrounds and rich information.

To rapidly detect adversarial RSIs and accurately restore them to their natural state with semantic consistency in a unified framework, we propose a diffusion-based dual-stage adversarial defense (DiffDual-AD) as a plugin for baseline DNN against various adversarial attacks. DiffDual-AD can overcome the limitations of the individual defense strategy mentioned above for RSI interpretation. Specifically, in the first stage of DiffDual-AD, we propose adversarial detection method based on the score expectation (AD-SE). The logarithmic probability density (i.e., score) with respect to an image indicates its momentum toward high-density regions of natural data [38]. Thus, during the forward process of the diffusion model, we propose to use multiview observations of a single-test RSI and estimate the score expectation to minimize the uncertainty. The maximum mean discrepancies (MMDs) between the score expectations of natural RSIs and the test RSI are then adopted as the detection statistic. Based on the result of the first stage, only the detected adversarial RSIs can activate the reverse process to restore them to the natural ones. We propose distance and label-guided adversarial purification (DL-GAP) as the second stage. DL-GAP can retain semantic consistency between purified RSIs and original RSIs by implementing two novel guidelines, specifically designed for RSIs with complex backgrounds and details. In DiffDual-AD, instead of using the denoising network for generative tasks, we employ a U-Net denoiser constrained for adversarial defense as the prior. During purification, our prior gradually removes adversarial perturbations and effectively expands the reverse-sampling search space, ensuring that the purified RSIs are mapped into the adversarial-free domain.

DiffDual-AD stands out for its capacity to accurately detect adversarial RSIs and purify them in an unsupervised way. The detect-and-purify strategy integrated into the diffusion model offers an effective pipeline to the challenge of preserving the utility of each RSI while ensuring a dual safeguard against malicious RSIs. DiffDual-AD is also computationally feasible for deployment, mainly because AD-SE requires low time cost, and DL-GAP only purifies the detected adversarial RSIs based on the results of AD-SE. Compared with other diffusion-based methods to purify all samples, DiffDual-AD takes a lower average computational cost with mixed samples. As scene classification task represents the most fundamental goal in RSI interpretation [20], [39], extensive robustness evaluation experiments are carried out on UC Merced (UCM) [40], aerial image dataset (AID) [41], and UAE-RS [42], three

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optical scene classification datasets. The results indicate that DiffDual-AD delivers satisfactory defense performance across six different DNNs with transparent-box and opaque-box attacks. The main contributions of this article are summarized as follows.

1) We propose a novel diffusion-based dual-stage adversarial defense framework, DiffDual-AD, which involves an adversarial detection stage followed by an adversarial purification stage. The unified and universal framework enhances the robustness of DNNs against various adversarial attacks in remote sensing, preserving potentially valuable adversarial RSIs without any information loss.
2) We propose the score expectation-based adversarial detection during the forward process, which employs a denoised prior designed for adversarial defense to estimate the expected score of RSIs. The MMDs computed between the expected scores of the natural and test RSIs can efficiently detect the adversarial content.
3) We propose incorporating the novel distance and label guidance into the reverse process to remove adversarial perturbations while maintaining semantic integrity of RSIs. The denoised prior ensures that the purified RSIs are restored to their natural state as closely as possible.
4) Our experiments on three RSI benchmarks validate DiffDual-AD's effectiveness against diverse adversarial attacks, which shows that DiffDual-AD can provide robust protection to DNNs trained with RSIs.

The rest of this article is organized as follows. A related work review is given in Section II. Section III elaborates on our proposed DiffDual-AD framework. Section IV presents the experimental results and discussions. Conclusions are finally summarized in Section V.

II. Related Work

A. Adversarial Attacks in Remote Sensing

Since Szegedy et al [43] demonstrated the existence of adversarial samples, ensuring robustness against various adversarial attacks has been crucial to modern DNNs. From the perspective of RSI interpretation like scene classification, given a natural sample $\boldsymbol{x}_0 \in \mathbb{R}^{h\times w\times c}$ with a true label $\boldsymbol{y} \in \mathcal{Y} = \{1, 2, \ldots, C\}$, an adversarial attack can generate malicious perturbation $\boldsymbol{\delta} \in \mathbb{R}^{h\times w\times c}$ to craft the adversarial sample $\boldsymbol{x}_{\text{adv}}$ by maximizing the loss function $\mathcal{L}_\theta$ of a well-trained DNN f_θ with its parameters θ, where h, w, and c represent the height, width, and number of channels

$$\boldsymbol{x}_{\text{adv}} = \boldsymbol{x}_0 + \arg\max_{\boldsymbol{\delta}\in\mathcal{A}} \mathcal{L}_\theta(\boldsymbol{x}_0 + \boldsymbol{\delta}, \boldsymbol{y}). \tag{1}$$

In the digital domain, the constraint $\mathcal{A}$ typically uses L_p, $p \in \{1, 2, \infty\}$ distance to limit $\boldsymbol{x}_{\text{adv}}$ visually indistinguishable from $\boldsymbol{x}_0$. With respect to the remote sensing field, many existing papers have developed preferable adversarial attacks on RSIs to indicate the vulnerability on different downstream remote sensing tasks, such as scene classification [13], [20], [44], [45], object detection [46], [47], semantic segmentation [48], and change detection [49], [50]. Specific attacks for multimodal models are also proposed to constrain the process of information fusion in both the spatial and frequency domains [51].

More seriously, adversarial RSIs have been realized in the physical domain, posing a significant and tangible threat to DNNs in remote sensing tasks. Currently, physical adversarial attacks are categorized primarily into altering the physical targets themselves [52], [53], [54] and their backgrounds [55], [56], [57]. For example, Du et al. [53] present an early attempt at physical attack on RSIs, optimizing and placing adversarial patches on cars to greatly impair the performance of object detectors. Chen et al. [56] find the digital-to-physical visual inconsistency problem and propose a novel harmonization-based attack in remote sensing scenes. Wang et al. [57] create innovative adversarial backgrounds and place them beneath the victim targets, which can effectively conceal the targets from detection. Therefore, it is imperative to conduct research on adversarial defense for RSI interpretation.

B. Adversarial Defense in Remote Sensing

Adversarial attacks highlight the vulnerabilities of DNN-based RSI interpretation systems, prompting concerns over their applications. Thus, robust defense strategies with RSIs are urgently desired to resist these threats and mitigate potential severe outcomes. AT-based defense remains a favorable choice for existing research in the remote sensing field. Su et al. [20] first introduced AT into the scene classification and demonstrated its defensive effectiveness for adversarial RSIs. Chan-Hon-Tong et al. [58] obtained a similar conclusion on semantic segmentation with RSIs. Zhang et al. [59] proposed a novel structured adversarial self-supervised pretraining paradigm for robust object detection in RSIs. Xu et al. [60] also combined AT with self-supervised learning to expand the training set with unlabeled data. However, these methods cannot be practical, as they heavily rely on the training data and necessitate downstream fine-tuning to cope with the continuously emerging unknown adversarial RSIs. Moreover, AT-based defenses are hard to restore adversarial RSIs.

In addition to AT in different versions, adversarial detection and purification advance the progress of adversarial defense in remote sensing. Chen et al. [26] proposed an adversarial detection method by obtaining the confidence soft threshold of each class for scene classification. Aldahdooh et al. [22] fused several statistical metrics for the detection of adversarial RSIs. In terms of adversarial purification, Zhou et al. [28] presented an adversarial noise-removing network (ARN) to eliminate adversarial noise by learning generalizable invariant features across attacks. Xu et al. [27] proposed a task-guided denoising network (TGDN) to align adversarial and natural RSIs for adversarial purification. Naseer et al. [30] introduced a GAN-driven architecture named neural representation purifier (NRP) that can be trained in a self-supervised manner, modeling the distribution of adversarial RSIs and restoring them to a natural state. Yu et al. [36] developed UAD-RS, a universal adversarial purification approach with pretrained diffusion models on different RSI datasets. Yet, the absence of adversarial detection limits the real-world application of UAD-RS by failing to predetect the potential adversarial RSIs before purification.

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Despite many efforts for adversarial defense in remote sensing, limitations still exist. Detection-only methods cannot produce inference results, and directly discarding the detected adversarial RSIs results in a significant loss of observational information. Performing a purification-only method on all mixed RSIs leads to high computational costs. To overcome these challenges, as the first dual-stage adversarial defense framework integrated into diffusion model in remote sensing, our proposed DiffDual-AD takes a rapid adversarial detection during the forward process and only purifies the detected malicious RSIs during the reverse process, thereby preserving all the observations and obtaining correct inference results for purified adversarial RSIs.

C. Adversarial Purification With Diffusion Models

Diffusion models [61], initially introduced as probabilistic generative models for unsupervised learning, can transform a complex data distribution into a relatively simpler one, such as the Gaussian distribution, and vice versa. Recently, they have been applied to adversarial purification for empirical [35], [37], [62], [63] and certified [64], [65] robustness. The unified procedure for diffusion models with adversarial purification involves two complementary Markov processes, the forward process and the reverse process. The forward process incrementally introduces Gaussian noise to an adversarial sample, ultimately converging into the Gaussian distribution. The reverse process uses a pretrained denoising network to reconstruct natural samples from the diffused ones by estimating the score function and reversing the Markov chain. Performing the forward-and-denoise processes can effectively mitigate imperceptible adversarial perturbations, and theoretically, the purified sample closely resembles a natural one with high probability [35], [62].

Although the reverse process of diffusion models has been developed for adversarial purification, the forward process remains underexploited for adversarial defense. Moreover, the reverse process involves sequentially denoising from timestep t to $t-1$, which requires excessive costs. In the field of RSI interpretation, it is impractical to purify every captured RSI in real time. Thus, we propose to use score expectations of natural and test RSIs during the forward process to predetect adversarial RSIs through MMD calculation.

Another critical aspect is the trade-off between maintaining semantic consistency without losing important local details and mapping the adversarial RSIs to the adversarial-free domain. DiffDual-AD introduces a distance guidance suitable for RSIs regarding that proximity, and also incorporated label guidance to highlight the high-probability label at the concluding steps of the purification. Our novel denoiser constrained for adversarial defense ensures that the adversarial perturbations are completely removed. The detect-and-purify strategy within the diffusion model forms our proposed DiffDual-AD framework for the protection of baseline DNNs trained with RSIs.

III. Proposed Method

A. Framework Overview

Fig. 2 illustrates the specific workflow of our proposed DiffDual-AD framework, which can be applied in front of different baseline DNNs as an image preprocessing plugin, with the goal of detecting and purifying adversarial RSIs into the natural ones. The first stage, adversarial detection part, is embedded in a forward process with a natural RSI set $\{\boldsymbol{x}_0^{(i)}\}_{i=1}^m$ and a test RSI $\tilde{\boldsymbol{x}}_0$. During forward process from $t=0$ to $t=T$, the score expectation estimated by a pretrained U-Net denoiser $\phi_\vartheta(\boldsymbol{x}_t, t)$ serves as a detection statistic to reveal the discrepancy between natural and adversarial distributions. To further measure the distance of the score expectations with both magnitude and direction information, we compute MMDs of the score expectation from the m natural RSIs $\{\boldsymbol{x}_0^{(i)}\}_{i=1}^m$ and the test RSI $\tilde{\boldsymbol{x}}_0$. The result is then compared with a threshold to determine if the test RSI is adversarial. The total time length of the forward process and the threshold are empirically set, denoted as T and ρ.

Based on the results of adversarial detection, the processing of $\tilde{\boldsymbol{x}}_0$ has the following two possible cases. If $\tilde{\boldsymbol{x}}_0$ is detected as a natural RSI, it is directly sent to the baseline DNN $f_\theta(\boldsymbol{x})$ for inference. If $\tilde{\boldsymbol{x}}_0$ is probably an adversarial RSI, the second stage, adversarial purification part, would iteratively denoise the diffused RSIs $\tilde{\boldsymbol{x}}_T$ to the adversarial-free domain.

By introducing Gaussian noise to RSIs, the forward process renders the pixel distributions of natural and adversarial RSIs indistinguishable, effectively submerging the adversarial perturbations into the Gaussian noise. During the reverse process from $t=T$ to $t=0$, we introduce a distance and label guidance, and derive a novel transitional distribution, aiming to maintain semantic information and local key features during purification. With Gaussian noise and adversarial perturbations simultaneously eliminated, the purified RSI $\hat{\boldsymbol{x}}_0$ is expected to be correctly predicted even by the attacked baseline DNN. Integrating adversarial detection and purification strategies into the two processes of the diffusion models, the DiffDual-AD framework can improve robustness against adversarial attack and ensure reliable inference for baseline DNNs under ideal conditions. More details are discussed below.

B. Adversarial Detection Based on Score Expectation

Adversarial detection manifests its value by quickly identifying whether the input is adversarial or not and discarding the former to safeguard system integrity. In remote sensing, due to the constant changes in natural and human activities on the Earth's surface, RSIs captured at a specific time and under specific conditions cannot be obtained again through repetitive operations. Thus, each RSI should be adequately interpreted to maximize its value. Directly filtering out adversarial RSIs may cause critical information loss and they need to be restored to natural ones through adversarial purification, which will be achieved by the reverse process of diffusion model. In order to quickly detect possible adversarial RSIs while preparing diffused RSIs for the second stage, we propose to embed adversarial detection into the forward process, using the score expectation that can be estimated by the pretrained denoiser as the detection statistic, named as AD-SE method.

Assuming that natural RSIs follow an unknown distribution $q_0(\boldsymbol{x}_0)$, the forward process can be denoted as $\{\boldsymbol{x}_t\}_{t\in[0,T]}$ with a total time length $T>0$ starting at $\boldsymbol{x}_0$, such that the diffused

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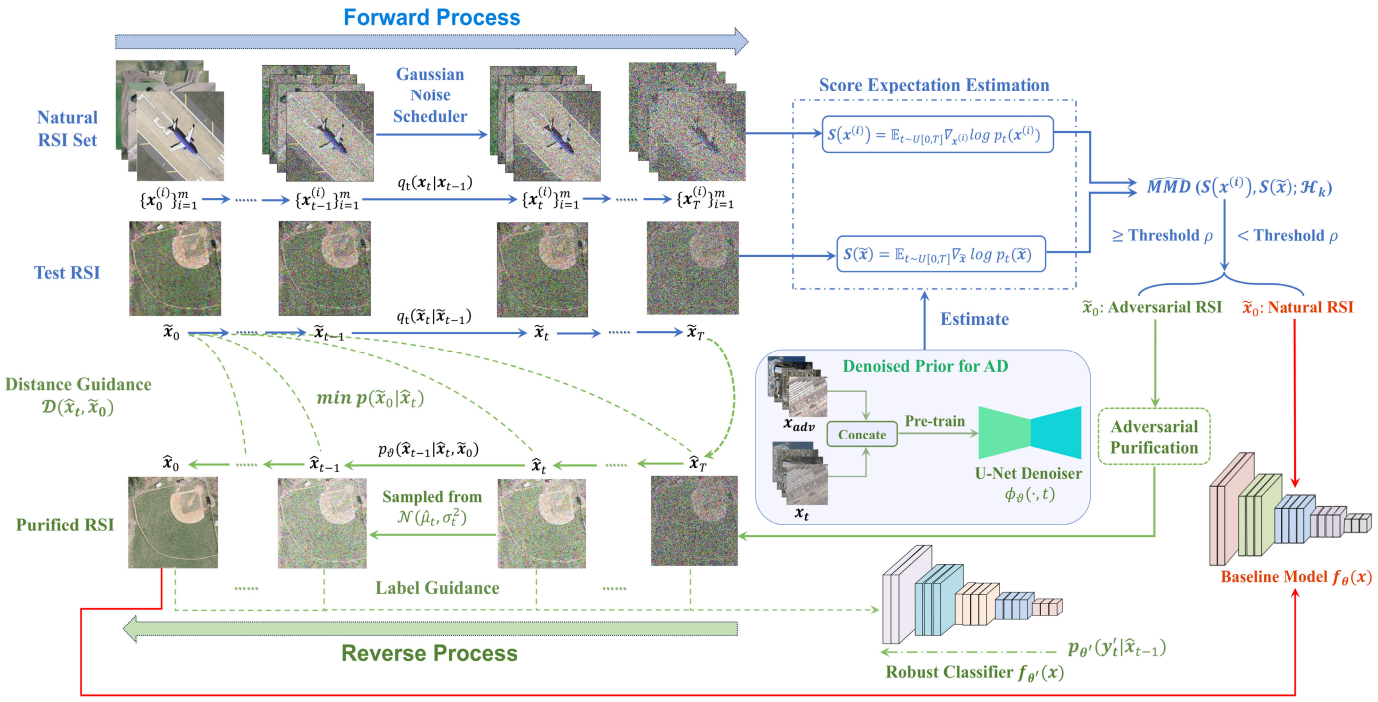


Fig. 2. Overview of the proposed DiffDual-AD framework, outlined with blue lines for adversarial detection stage, green solid lines for image flows in adversarial purification stage, green dashed lines for implementing two novel guidelines, and red lines for the classification with the attacked baseline DNN. The adversarial detection employs a forward process to estimate score expectations of the test RSI and a natural RSI set with a U-Net denoiser constrained for AD. The expectations are compared via MMD to a threshold, classifying the test RSI as adversarial or natural. If adversarial, the test RSI is purified with distance and label guidance, where a reverse process iteratively refines it toward authenticity, eliminating both Gaussian noise and adversarial perturbations without compromising semantic and structural integrity. Ultimately, the purified RSI can be correctly classified even by the attacked baseline DNN.

RSI $\boldsymbol{x}_t$ derived from $\boldsymbol{x}_0$ satisfies the following stochastic differential equation with positive time increments. For the test RSI $\tilde{\boldsymbol{x}}_0$, this process is identical

$$d\boldsymbol{x}_t = h(t)\boldsymbol{x}_t\, dt + g(t)d\overline{\boldsymbol{w}}_t, \quad \boldsymbol{x}_0 \sim q_0(\boldsymbol{x}_0). \tag{2}$$

In this article, we consider the denoising diffusion probabilistic model (DDPM) following [61] to comply with the discretized time step in our problem formulation, where $h(t) = -(1/2)\beta_t$ and $g(t) = (\beta_t)^{1/2}$ with a linear Gaussian noise schedule $\beta_t = \beta_{\min} + (\beta_{\max} - \beta_{\min})t/T$. $\overline{\boldsymbol{w}}_t$ is a standard Wiener process in (2). DDPM allows the incremental transformation of the input data distribution to approximate a Gaussian noise distribution through the transition kernel $q_t(\boldsymbol{x}_t|\boldsymbol{x}_{t-1})$ as follows:

$$\begin{aligned} q_t(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) &= \mathcal{N}\left(\boldsymbol{x}_t; \sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t \boldsymbol{I}\right) \\ &= \sqrt{1-\beta_t}\boldsymbol{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}. \end{aligned} \tag{3}$$

With $\alpha_t = 1-\beta_t$ and $\bar{\alpha}_t := \prod_{i=1}^{t} \alpha_i$, any diffused RSI from intermediate step $t \in \{1, 2, \ldots, T\}$ can be directly rewritten from $\boldsymbol{x}_0$ by leveraging the characteristics of Markov chains

$$q_t(\boldsymbol{x}_t \mid \boldsymbol{x}_0) = \mathcal{N}\left(\boldsymbol{x}_t; \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0, \left(1-\bar{\alpha}_t\right)\boldsymbol{I}\right). \tag{4}$$

Directly pretraining denoising networks designed for generative tasks in [61], rather than specifically for adversarial defense, is not reasonable. Therefore, we propose to improve the denoising network $\phi_\vartheta(\boldsymbol{x}_t, t)$ with strongly attacked RSIs $\boldsymbol{x}_{\text{adv}}$ and their corresponding natural diffused RSIs $\boldsymbol{x}_t$ by minimizing the following objective (5) as the **denoised prior constrained for adversarial defense**, where λ_t is a weight with respect to t. The natural diffused RSI $\boldsymbol{x}_t$ and its corresponding strongly attacked $\boldsymbol{x}_{\text{adv}}$ are fused by concatenating along the channel dimension in training. In practice, the noise prediction of $\phi_\vartheta(\boldsymbol{x}_t, t)$ can estimate the scaled score function at any time step t: $-(1-\bar{\alpha}_t)^{1/2}\nabla_{\boldsymbol{x}} \log q_t(\boldsymbol{x}_t)$ [38]. In this article, we use U-Net as the architecture of denoised prior

$$\mathbb{E}_{t\sim U(0,T),\boldsymbol{\epsilon}\sim\mathcal{N}(0,\boldsymbol{I})}\left[\lambda_t \|\boldsymbol{\epsilon} - \phi_\vartheta(\text{concat}(\boldsymbol{x}_t, \boldsymbol{x}_{\text{adv}}), t)\|^2\right]. \tag{5}$$

To achieve adversarial detection in the forward process, the score function of a test sample or its norm, estimated by $\phi_\vartheta(\boldsymbol{x}_t, t)$, can measure the discrepancy between natural and adversarial distributions. The score function with respect to a sample can indicate its momentum toward the high-density region of natural data [37]. Nevertheless, only using the score of a single sample falls short in assessing whether a sample exhibits adversarial behavior, because the insufficient information available for a single sample can lead to considerable fluctuating results, as demonstrated in [37]. This phenomenon can be more pronounced in RSIs with complex background and rich semantic content. To minimize the uncertainty from one sample, we propose to leverage the multiview observations of both natural and test samples, namely all the diffused samples from time step 0 to T, and denote their score expectation $\boldsymbol{S}(\boldsymbol{x})$ as the detection statistic

$$\boldsymbol{S}(\boldsymbol{x}) = \mathbb{E}_{t\sim U(0,T)}\nabla_{\boldsymbol{x}} \log q_t(\boldsymbol{x}_t). \tag{6}$$

In the forward process of DiffDual-AD, the score expectations of a natural RSI set $\{\boldsymbol{x}_0^{(i)}\}_{i=1}^m$ and a test RSI $\tilde{\boldsymbol{x}}_0$ estimated

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by our denoised prior are denoted as $\boldsymbol{S}(\boldsymbol{x}^{(i)})$ and $\boldsymbol{S}(\tilde{\boldsymbol{x}})$. To measure the distance between $\boldsymbol{S}(\boldsymbol{x}^{(i)})$ and $\boldsymbol{S}(\tilde{\boldsymbol{x}})$ with both magnitude and direction information, we characterize their difference by calculating MMD. The MMD is a robust statistical measure that evaluates the similarity of two sets of data without the knowledge of their specific probability density functions. The MMD quantifies the difference in moments between two unknown distributions, with a higher MMD indicating more differences. To determine whether the test RSI comes from a distribution of natural RSIs, we employ MMD to estimate a form of distance between two data sets $\mathbb{P}_X = \{\boldsymbol{x}^{(i)}\}_{i=1}^m$ and $\mathbb{Q}_Y = \{\tilde{\boldsymbol{x}}\}$ as follows:

$$\begin{aligned}\widehat{\mathrm{MMD}}[\mathbb{P}_X, \mathbb{Q}_Y; \mathcal{H}_k] &= \frac{1}{m^2}\sum_{i,j=1}^{m} k\big(\boldsymbol{S}(\boldsymbol{x}^{(i)}), \boldsymbol{S}(\boldsymbol{x}^{(j)})\big) \\ &\quad - \frac{2}{m}\sum_{i=1}^{m} k\big(\boldsymbol{S}(\boldsymbol{x}^{(i)}), \boldsymbol{S}(\tilde{\boldsymbol{x}})\big) + k\big(\boldsymbol{S}(\tilde{\boldsymbol{x}}), \boldsymbol{S}(\tilde{\boldsymbol{x}})\big) \end{aligned} \tag{7}$$

where k denotes the kernel of a reproducing kernel Hilbert space $\mathcal{H}_k$, exemplified by the Gaussian kernel $k(a, b) = \exp(-(\|a-b\|^2/2))$. For our approach, we use a deep kernel MMD, aligning with the methodology of [66]. With this metric, we can estimate a kind of distance between the test RSI and a collection of natural RSIs. The result of (7) is then compared with a predefined threshold ρ. If the MMD value exceeds ρ, it suggests that the test RSI $\tilde{\boldsymbol{x}}_0$ does not originate from the natural distribution $q_0(\boldsymbol{x}_0)$ and is likely to be adversarial, triggering the next adversarial purification stage. In contrast, if the MMD value is below ρ, $\tilde{\boldsymbol{x}}_0$ is deemed free from attack and processed to a downstream baseline DNN for further interpretation.

In addition to ρ, it is crucial to determine an appropriate total time length T, which can affect the outcome of both stages. Assuming that a test RSI has been attacked as $\tilde{\boldsymbol{x}}_0 = \boldsymbol{x}_{\text{adv}} = \boldsymbol{x}_0 + \boldsymbol{\delta}$, the forward process can represent its perturbed representations $\tilde{\boldsymbol{x}}_t$ directly sampled from $\tilde{\boldsymbol{x}}_0$ with a closed form as follows:

$$\begin{aligned} q_t(\tilde{\boldsymbol{x}}_t|\tilde{\boldsymbol{x}}_0) &= \mathcal{N}\Big(\tilde{\boldsymbol{x}}_t; \sqrt{\bar{\alpha}_t}(\boldsymbol{x}_0 + \boldsymbol{\delta}), \big(1-\bar{\alpha}_t\big)\boldsymbol{I}\Big) \\ &= \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0 + \sqrt{\bar{\alpha}_t}\boldsymbol{\delta} + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon} \\ &\approx \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\phi_\vartheta\big(\tilde{\boldsymbol{x}}_t, t\big). \end{aligned} \tag{8}$$

During the forward process, $\bar{\alpha}_t$ gradually decreases and $1-\bar{\alpha}_t$ correspondingly increases. Given the perceptually indistinguishable constraint, the adversarial perturbation $\boldsymbol{\delta}$ is subtle modifications added to $\boldsymbol{x}_0$. At the time step T, the Gaussian noise $(1-\bar{\alpha}_T)^{1/2}\boldsymbol{\epsilon}$ needs to be sufficiently large to submerge the adversarial perturbation component $(\bar{\alpha}_T)^{1/2}\boldsymbol{\delta}$. Our specific denoised prior has the tendency to restore the final noisy RSIs $\tilde{\boldsymbol{x}}_T$ to the adversaral-free domain in the reverse process. However, T cannot be arbitrarily large. Excessive adversarial diffused RSIs with the adversarial perturbations completely submerged by Gaussian noise during the diffusion process, will have the score expectations that are extremely close to that of natural RSIs, rendering them difficult to distinguish in the adversarial detection stage.

Because the output of the U-Net denoised prior $\phi_\vartheta(\tilde{\boldsymbol{x}}_t, t)$ trained with heavily attacked RSIs is the scaled score, it has been sensitive to adversarial perturbations distinct from Gaussian noise when calculating and comparing the MMD of score expectations. This allows our used denoiser to differentiate between diffused RSIs with and without adversarial perturbations during the forward process, thereby effectively enhancing the performance of AD-SE.

C. Distance and Label-Guided Adversarial Purification

Leveraging the exceptional generative capability of diffusion models in the reverse process, we propose a DL-GAP with the U-Net denoised prior constrained for adversarial defense to purify the detected adversarial RSIs. The purification process takes $\hat{\boldsymbol{x}}_T = \tilde{\boldsymbol{x}}_T$ as the starting point and proceeds assuming that each step $p_\vartheta(\hat{\boldsymbol{x}}_{t-1} \mid \hat{\boldsymbol{x}}_t)$ samples with a Gaussian distribution, continuing until the purification is complete at $\hat{\boldsymbol{x}}_0$. This Gaussian transition in the reverse process can be expressed with a learned mean $\boldsymbol{\mu}_\vartheta(\hat{\boldsymbol{x}}_t, t)$ and t-dependent variance σ_t^2

$$p_\vartheta\big(\hat{\boldsymbol{x}}_{t-1} \mid \hat{\boldsymbol{x}}_t\big) = \mathcal{N}\big(\hat{\boldsymbol{x}}_{t-1}; \boldsymbol{\mu}_\vartheta\big(\hat{\boldsymbol{x}}_t, t\big), \sigma_t^2\boldsymbol{I}\big). \tag{9}$$

The denoised prior predicts the noise $\phi_\vartheta(\hat{\boldsymbol{x}}_t, t)$ at the time step t, including adversarial perturbations and Gaussian noise. This enables the derivation of the mathematical expressions for $\boldsymbol{\mu}_\vartheta(\hat{\boldsymbol{x}}_t, t)$ and σ_t^2 as the following equations. Moreover, as previously mentioned, this predicted noise serves as an approximation to the scaled score $\nabla_x \log q_t(\boldsymbol{x}_t)$, thereby incorporating knowledge of image statistics into the purification process

$$\boldsymbol{\mu}_\vartheta\big(\hat{\boldsymbol{x}}_t, t\big) = \frac{1}{\sqrt{1-\beta_t}}\left(\hat{\boldsymbol{x}}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\phi_\vartheta\big(\hat{\boldsymbol{x}}_t, t\big)\right) \tag{10}$$

$$\sigma_t^2 = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t. \tag{11}$$

In light of the inherent challenges in the tasks with RSIs, characterized by high interclass similarity and intraclass variability, the RSIs generated during the reverse process must maintain semantic consistency and preserve key features when compared to the originals. However, as illustrated in Fig. 3, in scenarios where adversarial RSIs are present, the absence of proper guidance or incorrect guidance during the reverse process can lead to a significant deviation of the purified RSI from the original in terms of image semantics and features, resulting in wrong categories predicted by the baseline DNN.

Therefore, drawing from the optimal loss function proposed in [67] for image detail restoration to ensure the high similarity between the purified RSI and the original, the distance guidance $\mathcal{D}(\hat{\boldsymbol{x}}_t, \tilde{\boldsymbol{x}}_0)$ we utilized is given by

$$\mathcal{D}\big(\hat{\boldsymbol{x}}_t, \tilde{\boldsymbol{x}}_0\big) = \omega\big(1 - \text{MS-SSIM}\big(\hat{\boldsymbol{x}}_t, \tilde{\boldsymbol{x}}_0\big)\big) + (1-\omega)\mathcal{L}^{\ell_1} \tag{12}$$

where $\omega = 0.84$ is empirically set as [67]. MS-SSIM [68], which stands for multiscale structural similarity index measure, enables a quality assessment of restored RSIs that is

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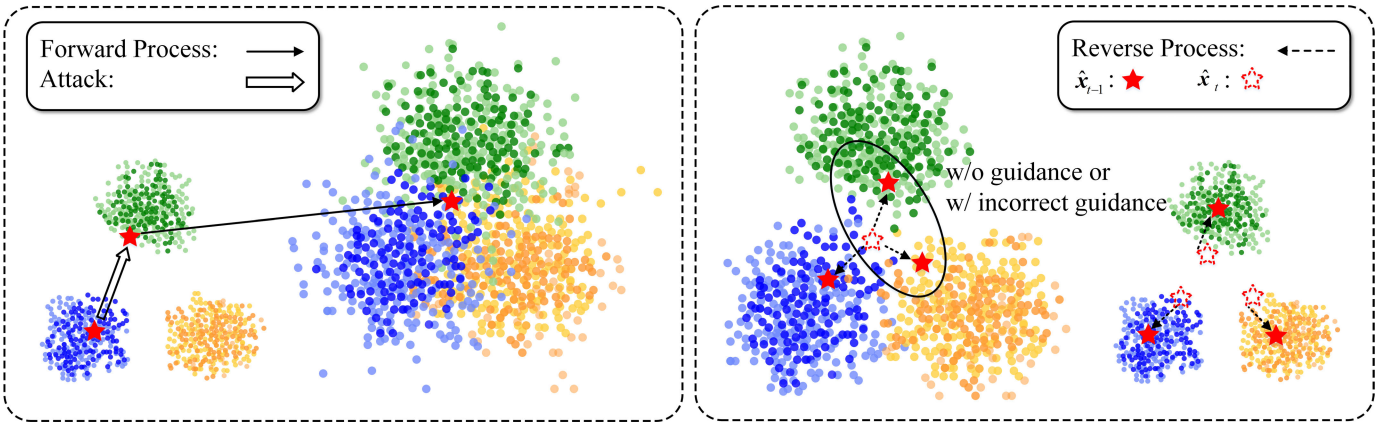


Fig. 3. Schematic of adversarial purification. The red star's trajectory denotes a test RSI that goes through the entire process of a pretrained denoiser based on DDPM, while the colored dots map diverse class distributions. In the presence of adversarial attack, without guidance or with incorrect guidance, the red star could move to a wrong distribution region, thereby compromising the robust accuracy of the baseline DNNs.

more closely aligned with subjective evaluations on multiple scales compared to the original. $\mathcal{L}^{\ell_1}$ normalizes the absolute differences of the corresponding pixel values between $\hat{\boldsymbol{x}}_t$ and $\tilde{\boldsymbol{x}}_0$. This distance guidance employs MS-SSIM to better preserve the structural and textural details in high-frequency regions, while decreasing the L_1 norm to maintain color contrast and luminance, where errors are weighted equally regardless of local structure. Compared to simple pixelwise comparisons like mean square error (mse), this distance provides a more comprehensive evaluation of image generation quality, considering the rich semantic content and local structural details within RSIs. However, excessive distance reduction may lead to the preservation of adversarial perturbations.

Our strategy of training the U-Net denoised prior $\phi_\vartheta(\hat{\boldsymbol{x}}_t, t)$ with a fusion of natural diffused RSIs and strongly attacked RSIs can help complete removal of adversarial perturbations. The attack intensity during the training should be strong enough to effectively expand the sampling search space for (9). Thus, our denoiser tends to overestimate the gap between the purified adversarial RSI and an adversarial-free domain, which facilitates the mapping of adversarial RSIs to the interior of the adversarial-free domain for the final result $\hat{\boldsymbol{x}}_0$. If we use the denoiser for generative tasks, there may be instances where residual adversarial perturbations remain in the purified RSIs. To more intuitively demonstrate our strategy of mapping purified RSIs into an adversarial-free domain, Fig. 4 presents a schematic comparison of the U-Net denoiser for generative tasks and ours designed for adversarial defense.

Another robust classifier $f_{\theta'}$ different from the baseline DNN is adversarially trained by minimizing the function with adversarial RSIs $\boldsymbol{x}_{\text{adv}} = \boldsymbol{x}_0 + \boldsymbol{\delta}$ as the objective (13), where γ is the attack perturbation intensity with L_∞ norm. It provides our proposed label guidance during the purification of adversarial RSIs, as an integral component concealed within the second stage of DiffDual-AD

$$\min_{\theta'} \mathbb{E}\left[\mathcal{L}(f_{\theta'}(\boldsymbol{x}_0), \boldsymbol{y}) + \max_{\|\boldsymbol{\delta}\|_\infty \le \gamma} \mathcal{L}(f_{\theta'}(\boldsymbol{x}_0), f_{\theta'}(\boldsymbol{x}_{\text{adv}}))\right]. \quad (13)$$

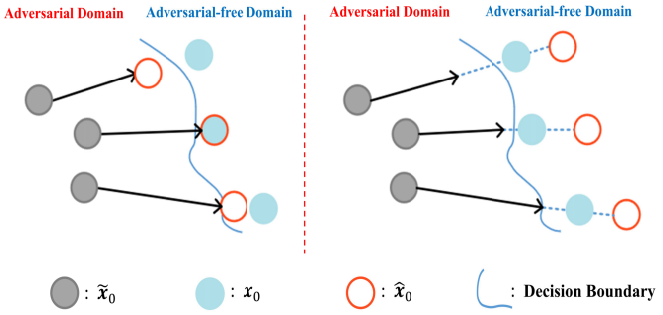


Fig. 4. Schematic comparison of adversarial perturbation elimination between (left) U-Net denoising network for generative tasks and (right) our denoiser prior for adversarial defense used in the reverse process.

The robust classifier $f_{\theta'}$ can make subsequent generated $\hat{\boldsymbol{x}}_t$ refine the features and details associated with the current output distribution $f_{\theta'}(\hat{\boldsymbol{x}}_t)$. At the beginning of adversarial purification, the purified RSI is submerged in Gaussian noise, leading to $f_{\theta'}(\hat{\boldsymbol{x}}_t)$ being quite uniformly distributed, and thus it cannot be reliable and the distance guidance (12) should predominate. When t approaches zero, adversarial perturbations are largely eliminated and the semantics of $\hat{\boldsymbol{x}}_t$ become clear. The value of true label in $f_{\theta'}(\hat{\boldsymbol{x}}_t)$ tends to obviously increase, so this guidance can reinforce the features and details consistent with the high-probability label at concluding steps of the purification process, providing reliable gradients back to the purified RSI and helping $\hat{\boldsymbol{x}}_0$ to be accurately classified, especially when the baseline DNN possesses network architectures similar to the robust classifier $f_{\theta'}$.

Specifically, we include the reverse transition distribution $p_\vartheta(\hat{\boldsymbol{x}}_{t-1} \mid \hat{\boldsymbol{x}}_t)$ with the original adversarial RSI $\tilde{\boldsymbol{x}}_0$ and output distribution of the robust classifier $\boldsymbol{y}'_t$, yielding a novel transitional distribution $p_{\vartheta,\theta'}(\hat{\boldsymbol{x}}_{t-1} \mid \hat{\boldsymbol{x}}_t, \tilde{\boldsymbol{x}}_0, \boldsymbol{y}'_t)$. Similar to the derivations from [34] and [62], the new transitional distribution has a positive correlation to the product of conditional probabilities $p_\vartheta(\hat{\boldsymbol{x}}_{t-1} \mid \hat{\boldsymbol{x}}_t)$, $p(\tilde{\boldsymbol{x}}_0 \mid \hat{\boldsymbol{x}}_t)$, and $p_{\theta'}(\boldsymbol{y}'_t \mid \hat{\boldsymbol{x}}_{t-1})$. To obtain this factorization of probabilities, we assume that $\boldsymbol{y}'_t$ and $\tilde{\boldsymbol{x}}_0$ are conditionally independent given $\hat{\boldsymbol{x}}_t$.

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$p_\vartheta(\hat{\boldsymbol{x}}_{t-1} \mid \hat{\boldsymbol{x}}_t)$ is characterized by the reverse process as (9), while $p(\tilde{\boldsymbol{x}}_0 \mid \hat{\boldsymbol{x}}_t)$ represents the probability of current RSI $\hat{\boldsymbol{x}}_t$ being purified into the original one $\tilde{\boldsymbol{x}}_0$. The higher this probability, the better the purification effect. Therefore, we approximate it in a heuristic formulation with the distance guidance $\mathcal{D}(\hat{\boldsymbol{x}}_t, \tilde{\boldsymbol{x}}_0)$ as follows:

$$p(\tilde{\boldsymbol{x}}_0 \mid \hat{\boldsymbol{x}}_t) = \exp\left(-s_t \mathcal{D}(\hat{\boldsymbol{x}}_t, \tilde{\boldsymbol{x}}_0)\right) \tag{14}$$

where s_t serves as a t-dependent weight that controls the magnitude of the distance guidance and we require $\mathcal{D}(\hat{\boldsymbol{x}}_t, \tilde{\boldsymbol{x}}_0)$ to be small to increase $p(\tilde{\boldsymbol{x}}_0 \mid \hat{\boldsymbol{x}}_t)$. As mentioned earlier, $\mathcal{D}(\hat{\boldsymbol{x}}_t, \tilde{\boldsymbol{x}}_0)$ encourages the semantic and structural proximity between $\hat{\boldsymbol{x}}_t$ and $\tilde{\boldsymbol{x}}_0$ throughout the reverse process. We take the logarithm of both sides and then calculate their gradients as $\nabla_{\hat{\boldsymbol{x}}_t} \log p(\tilde{\boldsymbol{x}}_0 \mid \hat{\boldsymbol{x}}_t) = -s_t \nabla_{\hat{\boldsymbol{x}}_t} \mathcal{D}(\hat{\boldsymbol{x}}_t, \tilde{\boldsymbol{x}}_0)$.

$p_{\theta'}(\boldsymbol{y}'_t \mid \hat{\boldsymbol{x}}_{t-1})$ is the label distribution of the current purified RSI of the robust classifier $f_{\theta'}$. We replace $\hat{\boldsymbol{x}}_{t-1}$ with $\hat{\boldsymbol{x}}_t$ for computational availability. The gradient of its logarithmic form as $\boldsymbol{g}' = \nabla_{\hat{\boldsymbol{x}}_t} \log p_{\theta'}(\boldsymbol{y}'_t \mid \hat{\boldsymbol{x}}_t)$ can be calculated by automatic differentiation through backpropagation. Then, we express the adjusted transitional distribution with distance and label guidance during the adversarial purification as follows:

$$\log p_{\vartheta,\theta'}(\hat{\boldsymbol{x}}_{t-1} \mid \hat{\boldsymbol{x}}_t, \tilde{\boldsymbol{x}}_0, \boldsymbol{y}'_t) \approx \log p(\boldsymbol{z}) + C \tag{15}$$

where C represents a constant and $\boldsymbol{z}$ follows

$$\boldsymbol{z} \sim \mathcal{N}\left(\boldsymbol{\mu}_\vartheta(\hat{\boldsymbol{x}}_t, t) + \sigma_t^2\left(\boldsymbol{g}' - s_t \nabla_{\hat{\boldsymbol{x}}_t} \mathcal{D}(\hat{\boldsymbol{x}}_t, \tilde{\boldsymbol{x}}_0)\right), \sigma_t^2 \boldsymbol{I}\right). \tag{16}$$

Therefore, by introducing the proposed distance and label guidance within the reverse process, the mean of sampling at each time step is translated by $\sigma_t^2(\boldsymbol{g}' - s_t \nabla_{\hat{\boldsymbol{x}}_t} \mathcal{D}(\hat{\boldsymbol{x}}_t, \tilde{\boldsymbol{x}}_0))$ based on $\boldsymbol{\mu}_\vartheta(\hat{\boldsymbol{x}}_t, t)$, with the variance $\sigma_t^2 \boldsymbol{I}$ remaining constant. The scaling factor s_t must be judiciously defined to positively influence the adversarial purification process. When t nears T, the adversarial perturbations within $\hat{\boldsymbol{x}}_t$ are largely submerged by Gaussian noise. Thus, distance guidance should take precedence and a greater magnitude of s_t is necessary to guarantee the restoration of semantic content and structural integrity for $\hat{\boldsymbol{x}}_t$. As t tends toward zero, the value of s_t must be adequately decreased to avoid an excessive level of uniformity that could otherwise result in the overimitation of $\tilde{\boldsymbol{x}}_0$, thereby retaining adversarial perturbations in $\hat{\boldsymbol{x}}_0$. Meanwhile, label guidance comes into play to enhance the features associated with a high-probability label as predicted by the robust classifier. In light of the analysis above and according to (8), we determine

$$s_t = \frac{3\sqrt{1-\bar{\alpha}_t}}{\gamma\sqrt{\bar{\alpha}_t}} \tag{17}$$

is directly proportional to the magnitude of Gaussian noise in $\tilde{\boldsymbol{x}}_t$, while disproportional to the magnitude of adversarial perturbation. Each dimension of Gaussian noise $(1-\bar{\alpha}_t)^{1/2}\boldsymbol{\epsilon}$ in (8) is approximately bounded by $[-3(1-\bar{\alpha}_t)^{1/2}, +3(1-\bar{\alpha}_t)^{1/2}]$, given that most samples from a standard Gaussian distribution lie within $[-3, +3]$. The adversarial perturbation component $(\bar{\alpha}_t)^{1/2}\boldsymbol{\delta}$ is constrained within the range $[-(\bar{\alpha}_t)^{1/2}\gamma, +(\bar{\alpha}_t)^{1/2}\gamma]$, as we limit its L_∞ norm to γ.

The proposed novel DL-GAP is capable of removing adversarial perturbations during the reverse process of diffusion models while maintaining semantic consistency with the original data and enhancing local key features associated with high-confidence labels. This ensures that the purified RSI $\hat{\boldsymbol{x}}_0$, even after the baseline DNN has been maliciously attacked, can still be accurately classified, achieving the trade-off between preserving semantics without losing important details and completely eliminating adversarial perturbations in the final purified RSIs.

Algorithm 1 DiffDual-AD

Input: m natural RSIs $\{\boldsymbol{x}_0^{(i)}\}_{i=1}^m$, a test RSI $\tilde{\boldsymbol{x}}_0$, a baseline DNN $f_\theta(\cdot)$, a robust model from AT $f_{\theta'}(\cdot)$, a U-Net denoised prior $\phi_\vartheta(\cdot,\cdot)$, attack perturbation intensity γ, hyper-parameter $\beta_{\max}$, $\beta_{\min}$, T, ρ.

Output: classification result from $f_\theta(\cdot)$ for the test RSI C.

Set initial time $t = 0$;
for $t = 1$ to T **do**
  $\beta_t = \beta_{min} + (\beta_{\max} - \beta_{\min})t/T$;
  $\bar{\alpha}_t := \prod_{i=1}^{t}(1-\beta_i)$;
  $\{\boldsymbol{x}_t^{(i)}\}_{i=1}^m = \sqrt{1-\beta_t}\{\boldsymbol{x}_{t-1}^{(i)}\}_{i=1}^m + \sqrt{\beta_t}\boldsymbol{\epsilon}$;
  $\tilde{\boldsymbol{x}}_t = \sqrt{1-\beta_t}\tilde{\boldsymbol{x}}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}$;
end for
Use $\phi_\vartheta(\cdot,\cdot)$ to estimate $\boldsymbol{S}(\tilde{\boldsymbol{x}}) = \mathbb{E}_{t\sim U(0,T)}\nabla_{\tilde{\boldsymbol{x}}} \log q_t(\tilde{\boldsymbol{x}}_t)$;
$\boldsymbol{S}(\boldsymbol{x}^{(i)}) = \frac{1}{m}\sum_{i=1}^m \mathbb{E}_{t\sim U(0,T)}\nabla_{\boldsymbol{x}} \log q_t(\boldsymbol{x}_t^{(i)})$;
Calculate MMD between two score expectations;
if MMD $< \rho$ **then**
  $\tilde{\boldsymbol{x}}_0$ is a natural RSI and $C = f_\theta(\tilde{\boldsymbol{x}}_0)$;
else
  $\tilde{\boldsymbol{x}}_0$ is an adversarial RSI to be purified;
  $\hat{\boldsymbol{x}}_t = \tilde{\boldsymbol{x}}_T$;
  for $t = T$ to 1 **do**
    $\mathcal{D}_t = \omega(1 - \text{MS-SSIM}(\hat{\boldsymbol{x}}_t, \tilde{\boldsymbol{x}}_0)) + (1-\omega)\mathcal{L}^{\ell_1}$;
    $s_t = \frac{3\sqrt{1-\bar{\alpha}_t}}{\gamma\sqrt{\bar{\alpha}_t}}$;
    $\boldsymbol{\mu}_\vartheta(\hat{\boldsymbol{x}}_t, t) = \frac{1}{\sqrt{1-\beta_t}}(\hat{\boldsymbol{x}}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\phi_\vartheta(\hat{\boldsymbol{x}}_t, t))$;
    $\sigma_t^2 = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$;
    $\boldsymbol{y}'_t = f_{\theta'}(\hat{\boldsymbol{x}}_t)$ and $\boldsymbol{g}' = \nabla_{\hat{\boldsymbol{x}}_t} \log p_{\theta'}(\boldsymbol{y}'_t \mid \hat{\boldsymbol{x}}_t)$;
    $\hat{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_\vartheta(\hat{\boldsymbol{x}}_t, t) + \sigma_t^2(\boldsymbol{g}' - s_t\nabla_{\hat{\boldsymbol{x}}_t}\mathcal{D}_t)$;
    $\hat{\boldsymbol{x}}_{t-1}$ sampled from $\mathcal{N}(\hat{\boldsymbol{\mu}}_t, \sigma_t^2\boldsymbol{I})$;
  end for
  $C = f_\theta(\hat{\boldsymbol{x}}_0)$;
end if
return C.

D. Complete Process of DiffDual-AD

In this article, we use score expectation and MMD calculation to distinguish natural RSIs from potential adversarial RSIs during the forward process of the diffusion model. We integrate distance and label guidance into the reverse process, gradually restoring the adversarial RSI back to the adversarial-free one while ensuring that their semantic, structural, and feature consistency remains consistent with the original RSI. Both stages are performed with the help of a specific U-Net denoised prior. The complete process of

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our proposed DiffDual-AD can be intuitively illustrated in Algorithm 1.

It should be noted that we simplify $\mathcal{D}(\hat{\boldsymbol{x}}_t, \tilde{\boldsymbol{x}}_0)$ as $\mathcal{D}_t$, and set the sampling mean $\hat{\boldsymbol{\mu}}_t$ at each time step during the adversarial purification process. The expression of $\hat{\boldsymbol{\mu}}_t$ can refer to step 22 in Algorithm 1.

IV. Experiments

A. Dataset Description

This section employs two optical RSI scene classification benchmarks, UCM [40] and AID [41]. Additionally, UAE-RS [42], the first opaque-box adversarial example dataset in the remote sensing community, is utilized for a more comprehensive defense evaluation. A brief introduction of each dataset is provided below.

1) *UCM Dataset:* The UCM dataset comprises 21 scene categories, with each scene featuring 100 RSIs. These images are of 256 × 256 pixels in dimension with a spatial resolution of 0.3 m/pixel. The dataset draws its imagery from the U.S. Geological Survey (USGS), representing a wide array of scenes from across the United States.
2) *AID:* Contrasting with the UCM's smaller scale imagery, AID presents a more expansive collection of 10 000 RSIs encompassing 30 scene categories. Each category includes between 200 and 420 images, each measuring 600 × 600 pixels.
3) *UAE-RS:* For the construction of the UAE-RS dataset, Xu and Ghamisi [42] utilized their proposed mixup and mixcut attack to generate adversarial RSIs by targeting the features in the shallow layers of multiple DNNs. The UAE-RS includes 1050 test adversarial RSIs from the UCM and 5000 ones from the AID for scene classification task. We refer to the two subsets in UAE-RS as UAE-RS UCM and UAE-RS AID. The ratio of the training set to the test set is 80%–20% for each dataset in the following experiments.

B. Experimental Settings

1) DNN Architectures: Six baseline models include AlexNet, ResNet-18, DenseNet-121, Inception-v3, RegNetX-400MF, and vision transformer (ViT-Base/16). The robust classier is a ResNet-18 trained with the corresponding attacked RSIs for the transparent-box setting and with the RSIs in UAE-RS for the opaque-box setting, while the training strategy is as (13). The classification results of baseline DNNs on the test sets are shown in Table I. Each result in Table I is averaged from ten independent experiments.

2) Attack Algorithms: Six prevalent adversarial attack algorithms are adopted in the transparent-box setting: FGSM [69], I-FGSM [70], PGD [70], Auto-PGD [71], C&W [72], and Deepfool [73]. FGSM adjusts the input data to maximize the objective function based on the backpropagated gradients. To improve the attack effectiveness of FGSM, I-FGSM, PGD, and Auto-PGD are improved versions of iterative gradient-based attack. The number of iterations for the iterative attacks is consistently limited to 5. C&W is an optimization-based attack with fixed hyperparameters $c = 0.1$ and $k = 20$. Deepfool can iteratively search for the minimum perturbation to make the natural RSIs cross the decision boundary.

TABLE I
Classification Accuracy (%) of Baseline DNNs on the Three Datasets

Model	UCM	AID	UAE-RS
AlexNet	91.23	91.20	24.56
ResNet-18	95.90	93.75	4.76
DenseNet-121	96.19	93.70	13.65
Inception-v3	95.33	92.50	24.17
RegNetX-400MF	96.76	94.40	23.26
ViT-Base/16	95.24	92.35	27.04

Transparent-box attacks generate adversarial RSIs based on the gradients and features extracted from the baseline DNNS. Conversely, opaque-box attacks are able to perturb RSIs without requiring knowledge of the networks. In our experiment, because the generation of UAE-RS is not targeted at the utilized baseline DNNs, we can directly use UAE-RS to evaluate the defensive effectiveness against opaque-box adversarial RSIs, which, in other words, constitutes the transfer attack.

3) Experimental Design: The comprehensive defensive evaluation of DiffDual-AD framework is divided into five main segments. The first part is to evaluate our proposed AD-SE after the forward process. Both adversarial samples attacked by a specific algorithm and their corresponding original ones are utilized, thus resulting in a dataset comprising equal parts of adversarial and natural RSIs in this part. We compare the detection performance of AD-SE with the state-of-the-art adversarial detection methods including feature squeezing (FS) [74], local intrinsic dimensionality (LID) [75], mahalanobis distance (MD) [76], kernel density estimation (KDE) [77], and ensemble adversarial detection (ENAD) [23] in a transparent-box setting. The perturbation intensity of gradient-based attacks is all fixed at $\gamma = 0.4$ with attack step size $\gamma/4$. We also present the adversarial detection results of comparative experiments on the UAE-RS. Given that adversarial detection becomes increasingly challenging with a reduced perturbation intensity and becomes relatively tractable with an augmented perturbation intensity, to broadly appraise the adversarial detection performances, we further compare AD-SE under different attack perturbation intensities $\gamma \in \{0.1, 0.2, \ldots, 0.6, 0.7\}$ against FGSM, I-FGSM, PGD, and Auto-PGD with other methods on the UCM dataset.

The adversarial purification poses significant challenges when conducted through unsupervised training, while most approaches require training based on specific adversarial perturbations. In the second part, we compare the performance of DL-GAP with four state-of-the-art defense methods against all the six transparent-box attacks. The four comparative methods include ARN [28], TGDN [27], NRP [30], and UAD-RS [36]. We also provide performances of the target networks without any defense. The perturbation intensity γ and step size of the gradient-based attacks are all set to 1 in this part. The third part demonstrates the adversarial robustness improvement with

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opaque-box adversarial RSIs, namely UAE-RS. The opaque-box defensive experiment is also conducted across six baseline DNNs to show the universal applicability of DiffDual-AD to different classification models.

In practical situations, once we become aware that adversaries have generated malicious RSIs against our baseline DNN, we activate the DiffDual-AD framework for defense. Therefore, to more closely emulate real-world scenarios, in the fourth section of experiments, we provide a complete assessment of DiffDual-AD, which includes the overall classification accuracy for the natural RSIs selected in AD-SE and the purified RSIs purified in DL-GAP, along with the response time and computational complexity of two stages. Finally, the fifth part provides some ablation studies.

4) Evaluation Metrics: To effectively differentiate the outcomes of the adversarial detection, we denote correctly detected adversarial and natural RSIs as true positives (TPs) and true negatives (TNs), respectively. Incorrectly detected adversarial and natural RSIs are termed as false negatives (FNs) and false positives (FPs), respectively. For the first stage, we employ the area under the receiver operating characteristic (AUROC) curve and area under the precision–recall (AUPR) curve as two metrics, which are consistent with the evaluation metrics in [23], [74], [75], [76], and [77].

1) *AUROC:* The area under the curve identified by specificity = TN/(TN + FP) and fall-out = FP/(TN + FP).
2) *AUPR:* The area under the curve identified by precision = TP/(TP + FP) and recall = TP/(TP + FN).

The defense accuracy (DA = $n_{\text{correct}}/n_{\text{total}}$) is adopted as the quantitative comparison metric of the adversarial purification part. n_{correct} denotes the number of RSIs that have been accurately classified, and n_{total} represents the size of the test dataset. A higher DA indicates a stronger defense.

5) DiffDual-AD Implementation Details: DiffDual-AD adopts U-Net as the backbone architecture of denoised prior $\phi_\vartheta(\cdot,\cdot)$ with the diffuser as detailed in [62], where $\beta_{\min} = 1 \times 10^{-4}$ and $\beta_{\max} = 2 \times 10^{-2}$. During training of the U-Net denoiser, strongly attacked RSIs from PGD at $\gamma = 0.8$ and their corresponding natural diffused RSIs are fused along the channel dimension with the objective (5) and $T = 500$ for 500 epochs per dataset. The pixels of each RSI in $\{0, 1, \ldots, 255\}$ are normalized to $[-1, 1]$. Adam is applied as the optimizer, with a learning rate of 2×10^{-4} and a dropout rate of 0.1. Every mini-batch in training contains 32 RSIs.

During the test time, the time step t is set to $\{5, 10, \ldots, 95, 100\}$ with $T = 100$. To ensure generalizability of the threshold ρ of AD-SE, we empirically optimize and evaluate it by cross-validation, selecting $\rho = 0.7$ with stable performances in the three used RSI datasets. The experiment software set-up comprises the Windows 10 operating system, the PyTorch 2.1.1 deep learning framework, and Python 3.10. The hardware configuration includes an Intel Core i7-10870H CPU and two NVIDIA GeForce RTX 3090 GPUs.

C. Adversarial Detection

Figs. 5 and 6 report the performance of FS, KDE, LID, MD, ENAD, and our proposed AD-SE against various transparent-box adversarial attack algorithms on the UCM and AID datasets. These results are based on the average values across six baseline network architectures. Our proposed AD-SE attains the highest score in 22 of the 24 scenarios, with a slight deficit in AUROC values when detecting RSIs attacked by C&W, as compared to the ENAD.

As observed, the AD-SE consistently achieves results above 90 across all gradient-based attacks, outperforming Deepfool and C&W by a narrow margin. Quantitative results indicate that the single-step FGSM, performing merely one iteration, excels in the rapid generation of adversarial RSIs yet is more discernible by adversarial detection. In contrast, I-FGSM, PGD, and Auto-PGD, through iterative gradient calculations, produce perturbations with smaller norms within the specified range, resulting in more subtle disturbances compared to FGSM. Among them, PGD and Auto-PGD can craft adversarial RSIs that pose a greater threat to baseline DNNs based on first-order gradients and random spherical projection. Thus, their adversarial detection outcomes are slightly inferior compared to those of I-FGSM. Deepfool identifies the shortest path by searching for and crossing a decision boundary based on orthogonal projection, and the optimization-based C&W produces the smallest perturbations among these attacks. Thus, detecting the adversarial RSIs generated by Deepfool and C&W can be more challenging, but AD-SE still demonstrates great performances. For instance, obvious improvement is observed on the AID with Deepfool. ENAD yields result of 82.01 and 81.88, our AD-SE can enhance the values to 86.97 and 87.06. The highest result is obtained by using AD-SE to detect FGSM on the UCM, with AUROC and AUPR values of 97.01 and 96.32, respectively. The general performance on the AID is slightly lower than that on the UCM, probably due to the smaller size with the more concentrated perturbations generated in the RSIs of UCM.

Given that the ENAD utilizes a fusion of statistical indicators derived from KDE, LID, and MD, it outperforms the stand-alone algorithms corresponding to these three statistical measures in detecting all types of adversarial RSIs, and its performance is the closest to that achieved by our proposed AD-SE. However, these comparative methods solely rely on the statistical indicators of single observation of a test RSI and pretrain simple classifiers to determine whether the test RSI is adversarial. Our proposed method utilizes multiview observations of a single-test RSI by progressively introducing Gaussian noise within a small diffusion time step, and estimates its expected score during the forward process. Benefiting from the superior performance of MMD in measuring two given distributions, MMD value is further calculated to differentiate between the expectations of natural RSIs and the test RSI in both magnitude and direction. Therefore, the AD-SE demonstrates the strongest adversarial detection performance, achieving the highest AUROC and AUPR across most DNNs and attack scenarios.

On the UAE-RS adversarial dataset, Fig. 7 illustrates the ROC and P-R curves for various adversarial detection methods applied to UAE-RS. In both the standard adversarial RSI dataset and the opaque-box scenario, the method proposed in this article continues to demonstrate superior performance. The red ROC curve of AD-SE is positioned closer to the top

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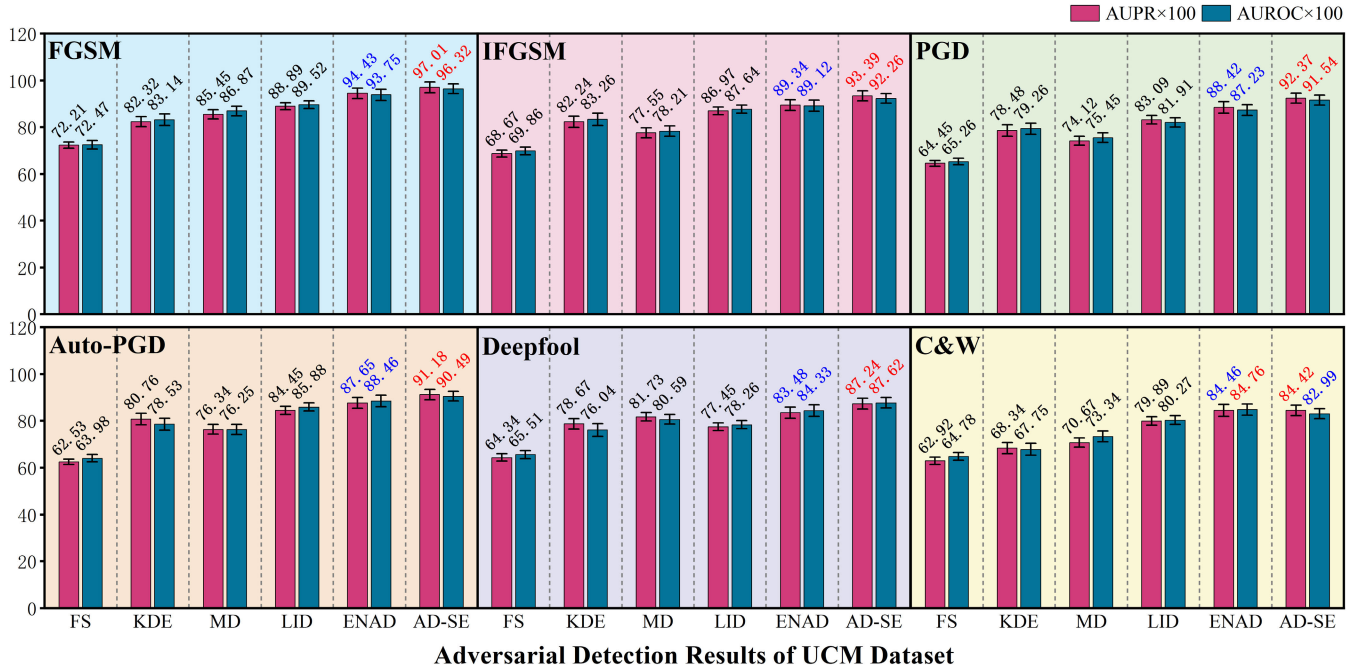


Fig. 5. Adversarial detection results on UCM with six types of transparent-box adversarial attacks. (Top row from left to right) FGSM, I-FGSM, and PGD. (Bottom row from left to right) Auto-PGD, Deepfool, and C&W.

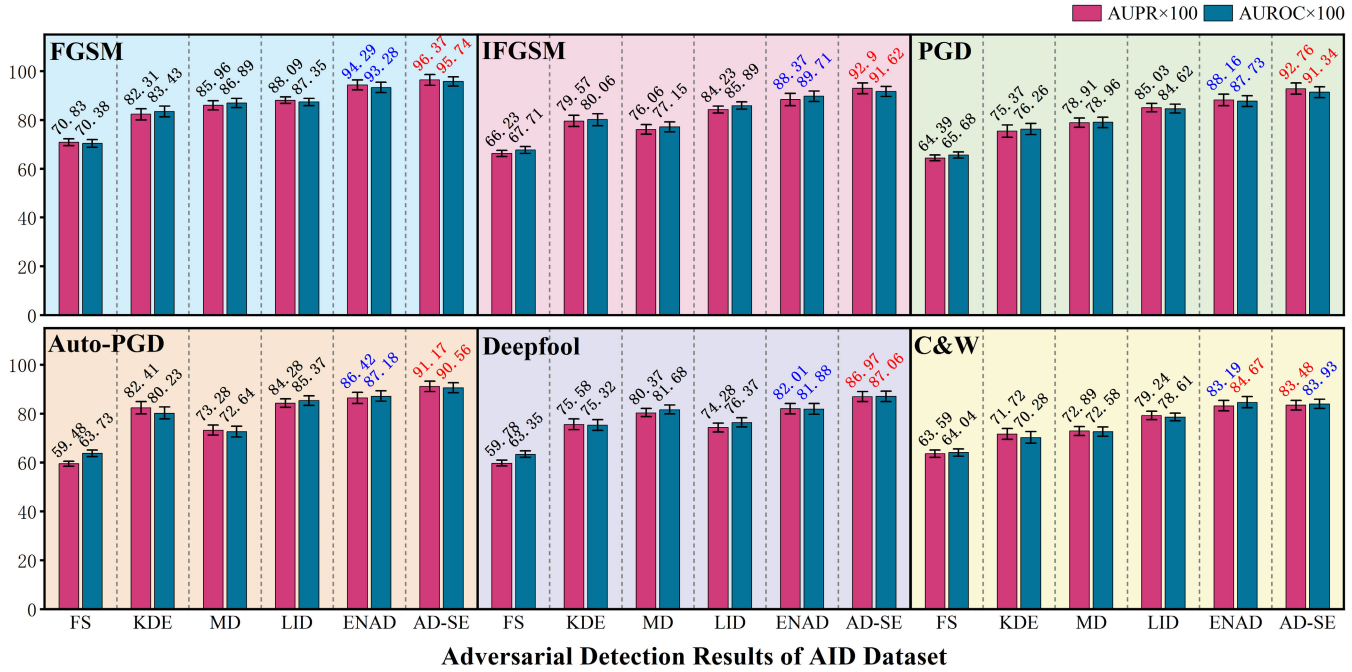


Fig. 6. Adversarial detection results on AID with six types of transparent-box adversarial attacks. (Top row from left to right) FGSM, I-FGSM, and PGD. (Bottom row from left to right) Auto-PGD, Deepfool, and C&W.

left corner, while the red P-R curve is closer to the top right corner, almost enveloping the curves of other methods. Given that the ROC curve is suitable for problems requiring a balance between positive and negative classes, while the P-R curve is particularly focused on the performance of the positive class, AD-SE proves robust under both metrics in various adversarial attack scenarios with both transparent-box and opaque-box settings.

In detecting adversarial RSIs under transparent-box conditions, we also evaluated the impact of varying attack perturbation intensities, denoted as $\gamma \in \{0.1, 0.2, \ldots, 0.6, 0.7\}$, on the performance of different adversarial detection methods across four gradient-based attacks. The results, presented as average AUROC $\times$ 100 on the UCM and AID datasets, are depicted in Fig. 8. It can be observed that distinguishing between adversarial and natural RSIs becomes increasingly challenging

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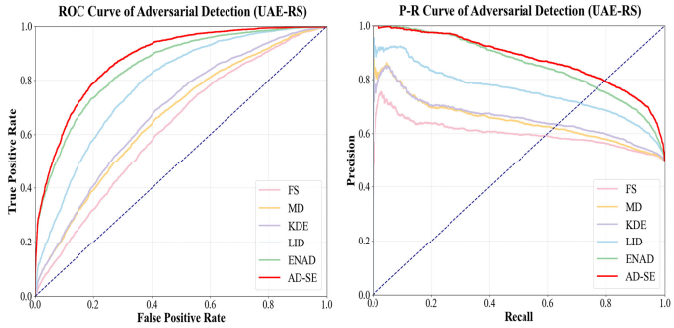


Fig. 7. Comparisons of ROC and P-R curve for evaluating different adversarial detection methods on the standard adversarial RSI set UAE-RS. The red curve represents ours.

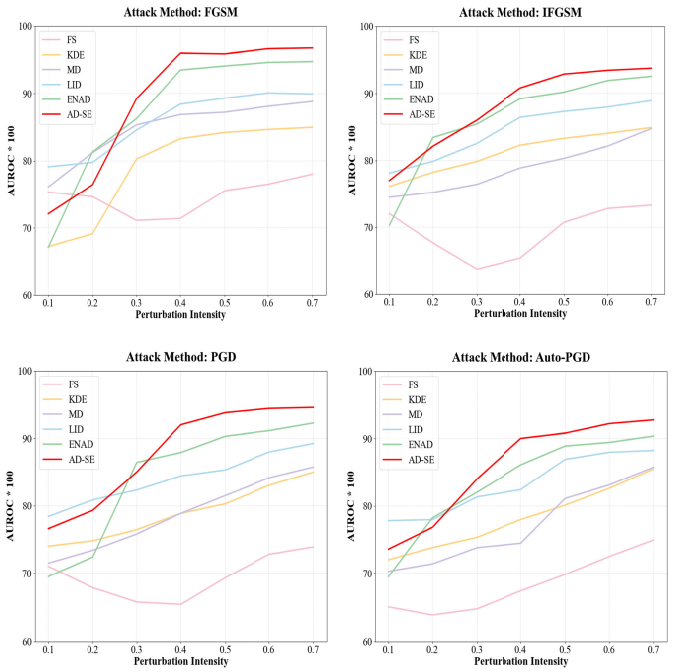


Fig. 8. Results of AUROC $\times$ 100 versus attack perturbation intensity of different adversarial detection methods against the gradient-based attacks. The red curve represents ours.

as γ decreases and becomes more tractable as γ increases, with the exception of the FS method. When $\gamma \leq 0.2$, our proposed method may not outperform other methods due to the insufficient perturbations in the adversarial RSIs, which are quickly overwhelmed by Gaussian noise during the forward process, causing too many multiview observations to resemble "natural diffused RSIs." Another possible reason may be that the attack intensity of the adversarial RSIs used for training the U-Net denoised prior was $\gamma = 0.8$, which is higher than $\gamma \leq 0.2$. As a result, the estimated score of adversarial RSIs with low attack intensity did not sufficiently differentiate from that of natural RSIs. However, it demonstrates a great advantage for $\gamma \geq 0.4$, with values mostly exceeding 90. This indicates that AD-SE is highly effective in detecting adversarial RSIs generated by strong attacks.

D. Adversarial Purification

1) Transparent-Box Scenarios: Regarding the second stage of adversarial purification, a comparison of our proposed DL-GAP for RSIs, with other state-of-the-art methods under both transparent-box and opaque-box attack scenarios, is conducted to evaluate their DA performances. Table II delineates the quantitative results of DL-GAP in contrast to its counterparts on the UCM and AID with transparent-box attacks. Each result in Table II is averaged from ten independent experiments. In Table II, there are a total of 72 transparent-box attacked scenarios that require robustness enhancement. The optimal values for each scenario are marked in red, and the suboptimal values are marked in blue.

When confronted with various transparent-box attacks, if no defenses are implemented, the DA on both datasets will not exceed 35%, severely compromising the normal functioning of the DNN models. The ARN exhibited consistent performance, defending against most attacks on the UCM with a DA from 50% to 60%, and on the AID with a DA from 40% to 50%. However, ARN eliminates the adversarial perturbations by only learning attack-invariant features, with the quality degradation of visual blurring and structural loss in the restored RSIs. Thus, our results suggest that the purification capability of ARN on RSIs may be constrained to a certain upper limit (only five suboptimal results for ARN). While the NRP demonstrated commendable robustness against FGSM attack, occasionally achieving optimal or suboptimal outcomes above 80% on the UCM, its efficacy notably deteriorated when contending with more sophisticated forms of adversarial perturbations, such as three types of iterative gradient-based attacks. The instability of training GANs, coupled with the complex background of RSIs, leads to the reduced performance of the NRP across more complicated attack (only four suboptimal and six optimal results for NRP, all from the FGSM attack).

The TGDN trains a denoising network using natural RSIs and corresponding I-FGSM attacked RSIs with three types of loss functions. Thus, the quantitative results also reveal that the DA against I-FGSM and even PGD sometimes achieves optimal on the AID and near-optimal on the UCM. However, akin to the limitations inherent in standard AT, the TGDN's defense is narrowly tailored to the threats it has encountered during training, lacking the robustness required to generalize against a broader spectrum of potential attacks. Similar to DL-GAP, the UAD-RS method utilizes the reverse process of a diffusion model to eliminate adversarial perturbations and the generated RSI quality has been substantially improved. It adaptively determines the optimal noise level, denoted as the total time length T in this article. However, the absence of guidance in the reverse process of UAD-RS may lead to less than satisfactory recovery of local structural and textual details in the purified RSIs, slightly impairing the baseline model's capacity to interpret semantic information within RSIs and resulting in classification errors. Nonetheless, the DA of UAD-RS is closest to that of our proposed method, achieving near-optimal or even optimal performance under the majority of attacks across both datasets (32 suboptimal and 15 optimal results for UAD-RS).

As observed in Table II, the DL-GAP manifests stable and superior performance, securing the premier DA across the vast majority of DNNs (21 suboptimal and 44 optimal

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TABLE II
DEFENSE ACCURACY (%) OF ADVERSARIAL PURIFICATION METHODS ON THE UCM AND AID WITH TRANSPARENT-BOX ATTACKS

Victim DNNs	Purification	Attack on UCM dataset						Attack on AID dataset					
		FGSM	IFGSM	PGD	Auto-PGD	C&W	Deepfool	FGSM	IFGSM	PGD	Auto-PGD	C&W	Deepfool
AlexNet	No Defense	10.95	0.00	30.48	19.67	0.00	3.57	12.80	0.00	24.10	18.45	0.00	2.90
	ARN [29]	61.33	59.62	56.67	53.33	59.81	47.38	49.55	47.85	39.30	38.85	46.45	43.20
	NRP [31]	68.81	36.95	35.05	29.29	36.76	29.52	71.45	31.15	27.20	28.55	29.25	27.35
	TGDN [28]	46.28	66.67	48.38	47.14	31.05	30.48	41.40	55.95	47.35	45.50	29.65	27.90
	UAD-RS [37]	65.43	63.33	64.86	57.86	61.19	55.48	65.80	57.85	58.55	49.80	58.40	52.35
	DL-GAP	69.52	66.43	69.29	61.90	57.38	53.33	67.45	62.35	60.70	58.85	55.40	49.80
ResNet-18	No Defense	33.90	0.00	35.33	5.48	1.14	2.57	25.70	0.00	34.50	3.65	0.00	1.45
	ARN [29]	58.76	57.33	58.95	51.19	58.09	48.33	48.35	45.30	38.40	41.15	48.55	45.45
	NRP [31]	82.86	32.86	67.03	58.33	40.19	49.52	78.25	19.65	35.60	30.45	20.55	19.10
	TGDN [28]	63.43	76.43	66.19	57.38	27.38	35.00	52.25	74.75	56.90	53.65	28.20	32.45
	UAD-RS [37]	75.48	69.48	75.52	61.43	66.86	52.24	66.55	59.45	66.15	58.60	61.35	49.85
	DL-GAP	84.29	79.67	82.24	73.05	70.57	68.95	74.30	75.95	77.10	73.95	66.35	61.40
DenseNet-121	No Defense	30.71	1.62	35.81	22.00	0.00	0.19	26.10	0.05	14.40	13.55	0.00	0.10
	ARN [29]	63.52	61.05	62.38	54.29	58.95	52.86	49.60	47.95	49.20	41.25	51.70	46.20
	NRP [31]	80.24	38.76	53.52	51.67	30.48	34.76	81.95	23.10	30.35	27.40	43.70	27.05
	TGDN [28]	68.57	73.81	61.43	49.29	44.05	33.81	50.10	73.40	53.25	44.95	37.60	33.10
	UAD-RS [37]	73.10	67.76	70.48	59.05	68.19	51.19	75.05	64.35	63.00	52.40	59.10	50.75
	DL-GAP	77.62	71.19	75.33	67.62	63.57	59.76	77.30	70.60	67.65	63.20	60.95	57.85
Inception-v3	No Defense	10.24	1.43	18.33	16.90	0.48	0.29	8.15	0.80	15.30	9.90	0.25	0.30
	ARN [29]	61.43	59.05	56.90	59.05	56.19	45.48	51.05	47.30	46.35	46.90	42.10	44.55
	NRP [31]	74.76	26.43	57.86	47.38	40.95	43.81	64.45	39.10	49.35	45.30	32.60	37.20
	TGDN [28]	52.62	70.62	64.95	52.62	34.76	28.33	49.55	68.00	54.65	49.80	40.15	33.60
	UAD-RS [37]	78.33	66.67	66.19	63.57	57.38	59.76	70.35	63.45	60.30	47.80	54.90	43.25
	DL-GAP	74.52	72.38	68.33	64.52	56.42	58.33	69.75	65.50	64.15	62.85	53.15	54.25
RegNetX-400MF	No Defense	34.24	0.02	29.68	16.43	0.00	7.24	32.75	0.00	28.45	11.25	0.15	5.25
	ARN [29]	58.76	57.33	57.38	50.71	47.62	50.24	42.50	41.30	44.40	38.55	53.70	47.35
	NRP [31]	81.42	44.48	55.90	49.05	49.90	44.76	80.45	15.80	38.50	36.75	35.55	29.60
	TGDN [28]	66.19	76.81	74.48	69.76	40.71	34.05	61.35	74.95	63.10	56.75	44.80	38.25
	UAD-RS [37]	82.48	69.10	75.24	66.19	75.81	56.90	76.90	67.70	66.45	55.55	62.75	58.80
	DL-GAP	74.29	77.38	81.90	72.86	66.19	63.57	75.80	73.70	72.00	70.25	64.20	62.65
ViT-Base/16	No Defense	31.67	0.24	23.33	15.05	0.43	0.57	30.60	0.55	18.75	16.50	0.75	1.65
	ARN [29]	50.96	50.24	51.90	52.86	49.76	44.05	45.45	44.75	42.10	41.85	47.25	37.00
	NRP [31]	79.29	35.48	43.81	43.10	29.29	36.43	76.85	29.60	42.75	41.80	32.25	39.00
	TGDN [28]	58.33	66.71	64.05	51.43	42.14	32.38	59.70	70.35	60.25	51.55	39.25	29.40
	UAD-RS [37]	80.24	68.95	68.33	58.10	48.81	54.52	69.40	58.55	61.35	64.20	45.25	53.70
	DL-GAP	79.76	68.62	71.57	65.24	61.67	56.43	75.65	64.85	66.50	60.65	55.20	56.05

results for DL-GAP, the best). Since the robust classifier is set to ResNet-18 in the experiment, the results of DL-GAP on ResNet-18 are particularly impressive. With specific distance guidance, DL-GAP enables the retention of detailed semantic information in RSIs, ensuring that their complex backgrounds are largely consistent with the originals. The robust classifier's output during the latter stage of purification facilitates backpropagation that emphasizes features corresponding to high-probability labels, making the purified RSIs more advantageous and compatible when encountering the DNNs with similar network architectures. Moreover, we can find that DL-GAP consistently maintains optimal performance under attack algorithms with iterative calculations such as Auto-PGD and I-FGSM. Despite often achieving optimal results, DL-GAP's numerical values remain relatively low when purifying C&W and Deepfool-generated adversarial RSIs, indicating a need for further investigation and improvement. DL-GAP greatly improves the usability and robustness of DNNs for RSI scene classification task, even under the highly challenging scenarios of transparent-box adversarial attacks (where a general threshold of 75% is considered for good usability, and can be reasonably reduced to around 70% when subjected to transparent-box adversarial attacks).

2) Opaque-Box Scenarios: In the opaque-box scenario, we utilize the standard adversarial RSI dataset, UAE-RS. It does not specifically target the baseline DNNs used in our experiments but rather takes advantage of the transferability of adversarial samples. Fig. 9 presents a visual comparison of the restoration effects of various adversarial purification methods on six examples from the UAE-RS dataset.

Upon close examination of Fig. 9, it is evident that the RSIs purified by the ARN method result in a loss of feature information, with many details appearing smoothed out, losing their original depth and texture. The image quality produced by GAN-based NRP seems somewhat degraded, as it appears to introduce additional noise aimed at disrupting adversarial patterns through its dynamic inference strategy. This strategy of blending the noise, rather than eliminating the original malicious noise, may be effective against certain types of adversarial perturbations but is not particularly effective against the adversarial RSIs from the UAE-RS. Although the TGDN managed to restore numerous background details, there remained some visually discernible residual perturbations and an inability to preserve texture details, which led to blurred regions. TGDN trains denoising DNNs only with I-FGSM attacked RSIs, potentially resulting in an inadequate generalization capability. This issue is particularly evident in the depiction of white rooftop and sea surface, as observed in the third and fifth columns of Fig. 9(e). Both UAD-RS and our proposed DL-GAP have produced commendable restorations

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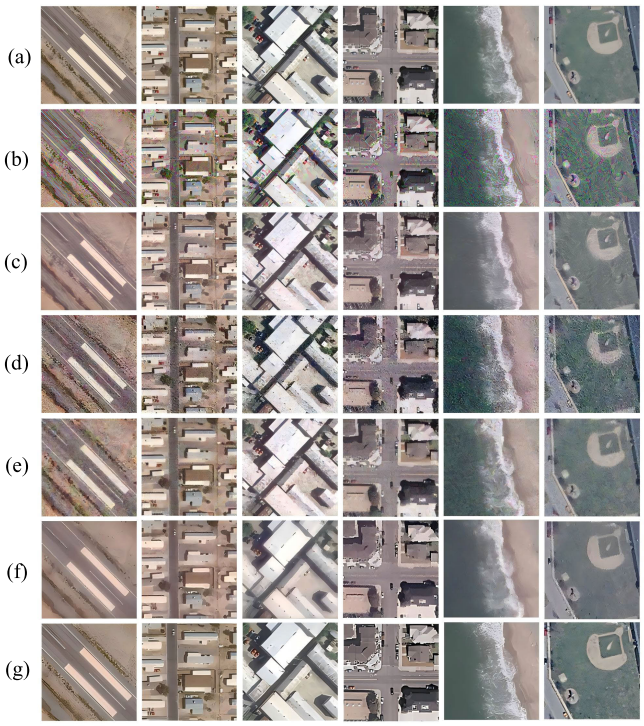


Fig. 9. Qualitative comparison for adversarial purification on the UAE-RS dataset. RSIs on the UAE-RS that are (a) adversarial samples and (b) ground truth. Purified RSIs on the UAE-RS from (c) ARN, (d) NRP, (e) TGDN, (f) UAD-RS, and (g) DL-GAP.

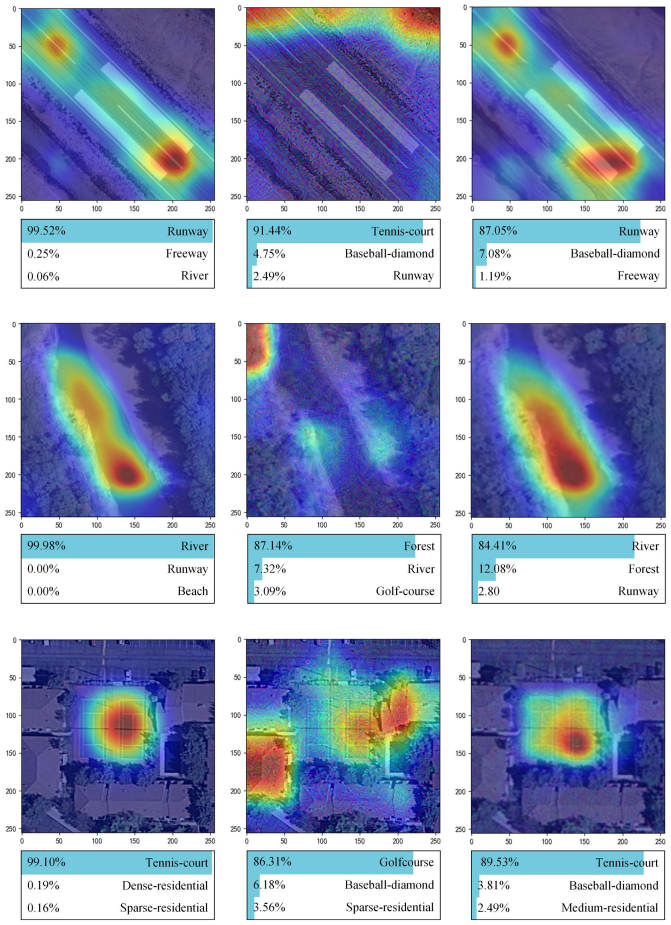


Fig. 10. Attention maps of three RSIs from the UAE-RS. The first column represents natural RSIs, the second column is the adversarial RSIs, and the third column represents the RSIs purified by DL-GAP. The blue bars at the bottom of each RSI show the top three predicted categories and probabilities.

of RSIs. However, UAD-RS's images exhibit slight differences in contrast and color compared to DL-GAP, due to the lack of specific distance guidance during the purification process. Additionally, UAD-RS falls short in restoring the sharpness of edges and lines that form contours, which DL-GAP handles more effectively. The proposed incorporation of distance and label guidance in DL-GAP significantly enhances the visual appeal of the purified RSIs. Moreover, due to the existence of the DDPM-based denoiser used for adversarial purification, the RSIs restored by DL-GAP no longer exhibit discernible patterns of adversarial perturbations, which are still slightly visible in ARN, NRP, and TGDN.

We selected three RSIs from the UAE-RS, and Fig. 10 visualizes the attention maps before the last fully connected layer of the baseline DNNs ResNet-18. Benefiting from the label guidance of the robust classifier within DL-GAP, the attention maps in the third column are refocused to the correct areas by the backpropagation of label guidance. For example, in the first row showing a runway, the adversarial images have completely undermined the inference logic of the baseline DNN, while the purified RSI realign attention back to the runway, leading to correct classification.

Quantitative evaluations were performed to evaluate the impact of various purification techniques on the UAE-RS dataset, as presented in Table III. Each result in Table III is averaged from ten independent experiments. Consistent with the practice of highlighting the best results in red and the second best in blue, the second-worst performance among all network architecture suggests that the target model of UAE-RS is akin to ResNet, aligning with the fact that the victim in [42] was ResNet-18. Corroborated by the qualitative findings of Fig. 9, residual adversarial perturbations in the purified RSIs of NRP and TGDN led to a significant reduction in their performance. Although ARN smoothed out some structural details in RSIs, it effectively removed most adversarial perturbations and even obtained the best result when using Inception-v3 for inference. In contrast, our proposed DL-GAP achieved the best results in five scenarios and the second best in one scenario, demonstrating its superior performance in the opaque-box attack scenario.

TABLE III

DEFENSE ACCURACY (%) OF ADVERSARIAL PURIFICATION METHODS ON THE UAE-RS IN A OPAQUE-BOX SETTING

Model \ Defense	ARN	NRP	TGDN	UAD-RS	DL-GAP
AlexNet	56.81	26.85	41.45	56.07	65.99
ResNet-18	21.36	12.00	26.36	60.70	64.97
DenseNet-121	48.04	33.78	46.83	63.76	70.03
Inception-v3	64.01	40.74	39.53	58.04	63.41
RegNetX-400MF	48.90	38.55	53.10	66.12	71.30
ViT-Base/16	43.71	27.36	53.27	57.79	62.16

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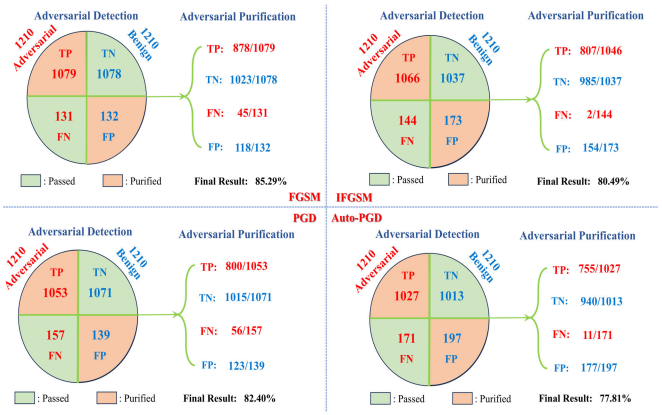


Fig. 11. Overall performance of DiffDual-AD framework against four transparent-box adversarial attacks on UCM and AID.

TABLE IV
COMPUTATIONAL COMPLEXITY COMPARISON ON UAE-RS

Test Data: 1050 adversarial RSIs from UAE-RS and 1050 natural RSIs

Methods \ Metrics	Time Cost (S/F)	Parameters (MB)	FLOPs
ENAD	0.632	4.15	91.47G
AD-SE	0.543	0	79.23G
TGDN	0.344	89.19	172.86G
UAD-RS	3.97	552,66	1.11T
DL-GAP	0.745	378.31	268.37G

E. Overall Performance of DiffDual-AD

To better simulate a practical scenario, adversaries craft adversarial RSIs targeting our deployed baseline DNN with transparent-box attacks, prompting the activation of our proposed DiffDual-AD framework as a protective strategy. Utilizing ResNet-18 as the baseline DNN, we attack the test sets of UCM and AID with FGSM, I-FGSM, PGD, and Auto-PGD. We randomly select half of the generated adversarial RSIs, blending them with their corresponding natural RSIs to form the dataset for this part of this experiment.

According to the operational mechanism of DiffDual-AD, when a baseline DNN equipped with DiffDual-AD encounters adversarial RSIs, the effective adversarial detection stage can initially identify them as TPs with high probability, followed by an adversarial purification process to eliminate adversarial noise. It is imperative to minimize the number of FNs. For natural RSIs, AD-SE allows them to proceed to baseline DNN, which then achieves an accurate classification with high benign accuracy as demonstrated in Table I. Even if misidentified as FPs, the natural RSIs, after undergoing the adversarial purification stage, are likely to be correctly classified with a high degree of accuracy.

Fig. 11 illustrates the overall performance of the DiffDual-AD system against four types of adversarial attacks. In each subplot, the circle on the left is divided into four quadrants, with the numbers in each quadrant indicating the RSIs categorized into that section during the adversarial detection. On the right side of each subplot, the classification results of the baseline DNN for the RSIs in these four quadrants are presented (correct/total), respectively. The **final result** is derived by summing the correct classifications across all four quadrants and dividing by the total number of natural and adversarial RSIs. We can observe that when faced with multiple types of transparent-box popluar attacks, the baseline DNN, under the protection of DiffDual-AD, achieves an average adversarial detection accuracy of 87.02% with AD-SE and an average final result of 81.50% with the entire DiffDual-AD framework. This is attributed to the synergistic operation of the two stages of DiffDual-AD, both of which are indispensable. The adversarial detection component efficiently and accurately distinguishes between natural and adversarial RSIs, while the adversarial purification component removes the adversarial perturbations in the detected adversarial RSIs. This two-stage defense strategy provides a dual safeguard for the baseline DNN, theoretically allowing only a minimal number of FNs to penetrate. Moreover, DiffDual-AD can serve as an image preprocessing plugin that can be readily deployed to protect the baseline DNN when facing adversarial attacks, and it remains inactive under normal circumstances.

Table IV compares the computational complexity of the proposed AD-SE and DL-GAP in DiffDual-AD with other methods on UAE-RS dataset. To more realistically simulate practical scenarios, we mixed 1050 adversarial RSIs from UAE-RS with 1050 natural RSIs, forming a test data of 2100 RSIs. The time cost was determined by averaging the processing time across 2100 RSIs, while computational complexity was assessed based on a standard image size of $256 \times 256 \times 3$ pixels. The top section of Table IV is dedicated to adversarial detection, while the bottom section is for adversarial purification. For adversarial detection part, we only compare AD-SE with ENAD, because according to the previous experiments, the performances of AD-SE and ENAD in detecting adversarial RSIs are significantly better than other methods. AD-SE involves the addition of Gaussian noise and some simple calculations, does not incur a significant time cost or FLOPs, and since it is in an unsupervised form, no parameters are saved to the device. As shown in Table IV, AD-SE is overall superior to ENAD across the complexity comparison.

For the adversarial purification, the reverse process of the diffusion model, which involves denoising a test RSI sequentially from time step t to $t-1$, requires our proposed DL-GAP to expend more time than TGDN. TGDN does not employ a diffusion model and only trains a denoising DNN for purification, resulting in reduced time expenditure. The U-Net denoiser in our DL-GAP occupies larger storage space than TGDN. However, the purification effectiveness of TGDN on adversarial RSIs is inferior to that of our proposed DL-GAP. It is worth noting that DL-GAP requires less time cost than diffusion-based UAD-RS. Compared with UAD-RS to purify all RSIs, DL-GAP only purifies the adversarial RSIs detected by AD-SE and achieves effective purification

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TABLE V
EFFECT OF T ON DEFENSE PERFORMANCE WITH UAE-RS

Metric \ Total-time	5	10	20	50	100	200
AUROC×100	63.68	75.74	84.03	**88.07**	85.28	79.69
AUPR×100	62.32	76.26	82.86	**88.11**	86.01	78.92
DA (%)	29.90	38.19	53.52	63.43	**70.03**	65.14
Final Result (%)	24.75	36.82	49.92	62.93	**65.57**	61.40

with a smaller number of time steps $T = 100$ for fewer reverse steps. We recognize that the computational time and resource consumption of DL-GAP present areas for possible improvement, which will be further discussed later.

F. Ablation Experiments

1) Hyperparameter Sensitivity Study: To investigate the impact of the total time length T of the diffusion model on the performance of AD-SE, DL-GAP, and the final result of DiffDual-AD, we conducted experiments on the UAE-RS dataset using DenseNet-121 with $T = 5, 10, 20, 50, 100, 200$. From the results in Table V, which are averaged from ten independent experiments, it is evident that when $T > 20$, AD-SE can achieve a usable AUROC and AUPR in detecting opaque-box adversarial RSIs, thus demonstrating the insensitivity of T for AD-SE. When $T = 50$, both AUROC and AUPR reach their optimal values, with 89.48 and 89.11, respectively, but the DA and final result after DL-GAP purification peaks at $T = 100$ with 70.03%. When $T = 200$, all the performances in Table V have dropped a lot. The variations in DA and final result are more pronounced compared to AUROC and AUPR, hence we prioritize achieving the optimal classification results, selecting $T = 100$ within our diffusion model. Furthermore, two purified RSIs from UAE-RS and their diffused RSIs after the forward process with different T are presented as in Fig. 12. When T is set to a low value, the purified RSIs tend to retain visually discernible residual adversarial perturbations, resulting in a low DA. Conversely, excessively increasing T, such as setting $T = 200$, can cause excessive blurring of details and oversmoothing within the purified RSIs.

2) Guidance Ablation Study: Table VI reports the contributions of two types of guidance utilized in DL-GAP to performance, with the corresponding attacks for UCM and AID being PGD. Each result in Table VI is averaged from ten independent experiments. The baseline DNN is ResNet-18. The top three rows of Table VI represent natural RSIs, while the bottom three rows demonstrate adversarial RSIs. The results indicate that both types of guidance can positively enhance the performance and effectively improve the defense accuracy of the baseline DNN against adversarial RSIs. The presence of distance guidance during the reverse process significantly enhances the classification accuracy of purified RSIs, while label guidance further improves the purification efficacy of DL-GAP. For instance, when testing the UCM dataset generated after a PGD attack on the baseline DNN, the baseline DNN achieved only 35.33% accuracy. After purification

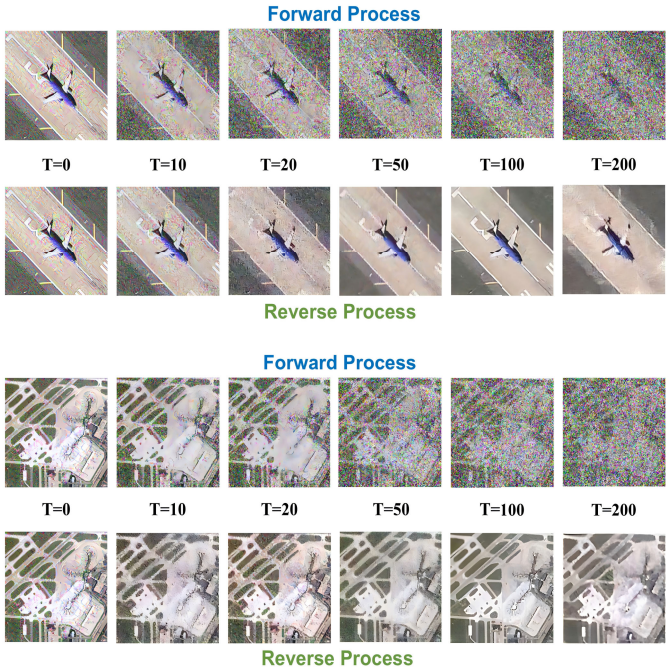


Fig. 12. Two examples in UAE-RS after DL-GAP adversarial purification with different total time length.

TABLE VI
PERFORMANCE CONTRIBUTION OF GUIDANCE IN DL-GAP

Attack	Distance	Label	UCM (%)	AID (%)	UAE-RS (%)
			95.90	93.75	–
	✓		92.14	89.05	–
	✓	✓	91.43	89.70	–
✓			35.33	14.40	4.76
✓	✓		78.09	67.20	61.24
✓	✓	✓	81.24	69.90	64.97

using DL-GAP with distance guidance alone, the classification accuracy rapidly increased to 78.09%. Subsequently, incorporating label guidance further elevated the results to 81.24%, reaching a level suitable for practical deployment. Moreover, DL-GAP, even if inadvertently applied to natural RSIs, does not substantially compromise the results, with accuracy still around 90%.

3) Denoiser Ablation Study: The DiffDual-AD does not indiscriminately employ natural diffused RSIs to train the U-Net denoising network; instead, it utilizes a fusion of RSIs subjected to strong adversarial perturbations and natural diffused RSIs (containing only Gaussian noise) to train the U-Net denoised prior. Our specific denoiser for adversarial defense exhibits heightened sensitivity to diffused RSIs with adversarial perturbations in estimating the score expectation. Moreover, it is capable of expanding the sampling search space during the DL-GAP purification, ensuring that both adversarial and Gaussian noise are removed after the reverse process. To illustrate the impact of employing our denoised prior ("Ours") versus a standard denoiser ("Normal") for

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generative tasks on the two stages of DiffDual-AD, Fig. 13 presents the comparative performance of AD-SE (AUROC × 100) and DL-GAP (DA) under transparent-box (PGD, UCM) and opaque-box (UAE-RS) scenarios, with the baseline DNN set as ResNet-18. γ represents the attack intensity to generate adversarial RSIs, which is used to train the denoiser. From the results in Fig. 13, it is evident that the denoised prior trained with a fusion of RSIs attacked by stronger adversarial perturbations ($\gamma = 1$) and natural diffused RSIs outperforms in both stages and scenarios, confirming the previously mentioned enhancement effects in AD-SE and DL-GAP.

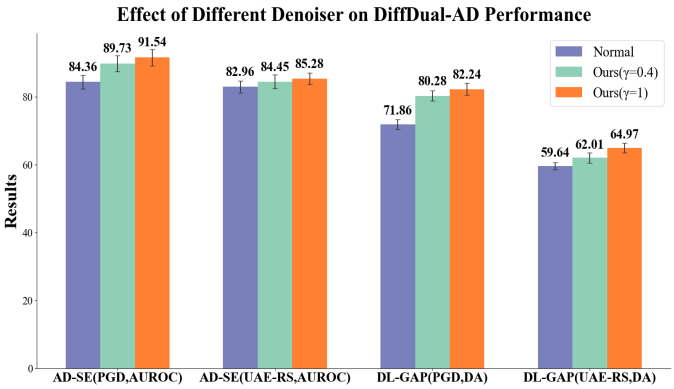


Fig. 13. Effect of different denoiser on DiffDual-AD performances (normal denoiser is designed for generative tasks; ours is trained with adversarial and natural diffused RSIs).

G. Discussion

1) Unsupervised Adversarial Detection: The first stage of DiffDual-AD, namely our proposed AD-SE method, can be considered unsupervised. Unlike MD, LID, and ENAD, which train a binary classifier using clean data and adversarial data labeled with "0" and "1," we directly calculate the MMD of score expectation for multiview observations from natural samples with that of the test sample after the forward process of diffusion model. Moreover, previous supervised adversarial detection methods train a customized detector for specific attacks or a particular baseline classifier, which often neglects the modeling of data distributions, resulting in limited performance against unknown attacks [22]. Experiments demonstrate that this novel statistical approach can be more effective than conventional methods in assessing the distributional discrepancy to detect adversarial RSIs in most scenarios. Given the complex backgrounds and rich features in RSIs, this statistical metric of multiple observations and score comparison proves to be reliable against both transparent-box and opaque-box adversarial attacks, even in an unsupervised form.

However, when detecting transparent-box adversarial RSIs at low attack intensity, previous experimental results indicated that the performance improvement was less obvious. To address this issue, one potential approach is to fine-tune our U-Net denoised prior with some weakly attacked RSIs fused with the corresponding natural diffused RSIs. Another approach could be to reduce the total time length T, since adversarial RSIs at low attack intensity are quickly overwhelmed by Gaussian noise during the forward process. This adjustment would increase the proportion of samples containing undiluted adversarial perturbations during the forward process.

2) Possible Improvement for DL-GAP: First, in contrast to conventional DNN-based purification algorithms, DL-GAP may require a longer period to purify adversarial RSIs due to its step-by-step reverse process. However, the rapid first stage during the forward process has already indirectly expedited the framework's operation by screening the adversarial RSIs that require purification of DL-GAP, which would otherwise necessitate the purification of every real-time captured RSI. Naturally, to further accelerate the reverse process of the diffusion model itself for better practical deployment, we can employ an efficient sampling strategy [62] or a sparse diffusion time step [38] for potential enhancement.

3) Extended Applications: This article conducts experiments on the scene classification task with optical RSIs; however, there are other remote sensing tasks such as object detection and semantic segmentation. In addition to these, RSIs also encompass SAR and infrared data. Nevertheless, DiffDual-AD could be a viable solution for adversarial defense in the remote sensing community.

V. Conclusion

In this article, an effective defense framework DiffDual-AD is designed to defend against various adversarial attacks on DNNs used for the automatic interpretation of RSIs. In light of the potential inclusion of significant information within each captured RSI, the rapid detection of adversarial RSIs and their restoration to clean ones, while preserving semantic consistency and local features, holds significant importance.

To achieve the goals, leveraging the framework of the diffusion model with an improved U-Net denoising network constrained for adversarial defense, we integrate adversarial detection and purification into the forward process and reverse process, respectively, forming a dual-stage adversarial defense method, DiffDual-AD. In the forward process, AD-SE has been developed. To deal with the complex background in RSIs and minimize the uncertainty from a test RSI, we propose to leverage the multiview observations of an RSI, namely all the diffused RSIs, and calculate their score expectations. By comparing the MMDs between the score expectations of natural RSIs and the test RSI, we can determine whether the test one is adversarial or not. In the second stage of the reverse process, we introduce the DL-GAP method and formulate a novel transitional distribution. The DL-GAP effectively eliminates adversarial perturbations from the adversarial RSIs while maintaining semantic information and key features, resulting in clear and nonadversarial RSIs.

Extensive experiments on three benchmark datasets demonstrate the effectiveness of AD-SE, DL-GAP, and overall performance in both transparent-box and opaque-box settings. In the opaque-box adversarial RSI dataset UAE-RS, the average AUROC reaches 88.07 and the average AUPR reaches 88.11. After adversarial purification with DL-GAP, the entire DiffDual-AD achieves an average final classification accuracy of 81.50% in the transparent-box scenario and the highest

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classification accuracy of 71.30% for the RSIs from UAE-RS. These performances surpass those of other advanced adversarial detection and purification algorithms when applied to adversarial RSIs.

In future work, we will address the limitations observed in the experiments of DiffDual-AD. For instance, the AD-SE demonstrates suboptimal performance in detecting adversarial RSIs generated at low attack intensities, as well as accelerating and reducing the complexity of the DL-GAP process. In addition, further exploratory enhancements could be pursued in other tasks such as object detection and semantic segmentation. Despite having slight flaws, the dual-stage adversarial defense framework presented in this article already offers a potential solution to significantly enhance the robustness of DNNs used in the field of remote sensing.

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